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# Contrastive learning-based multi-scale neural processing for thematic segmentation of lecture video transcripts

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**Abstract**

Transcripts of lecture videos may have topic transitions that are subtle and gradual, making segmentation difficult for computers. Current segmentation methods face limitations: lexical cohesion methods rely on overlap between words on the surface level without capturing deep meaning, transformer methods break a single long-form transcript into chunks to impair coherence, and methods based on structure cannot handle the casual nature of a transcript from a lecture video. This work introduces the Thematic Segmentation of Lecture Video Transcripts, a four-component hybrid technique involving deep semantic encoding in a long-context semantics model, neural segmentation on multiple scales using contrastive learning, graph refinement, and reinforcement learning using quality-driven rewards. Our experiments on the Synthetic Lecture Transcript Benchmark (210 transcripts) as well as on several real-world lecture datasets demonstrate that this proposed method achieves $F1 = 0.78 \pm 0.01$, $Pk = 0.27 \pm 0.01$, $WD = 0.25 \pm 0.01$, outperforming the evaluated baseline methods by an average of +10.4% relative F1 improvement (range +8.5% to +11.8%) over the strongest prior method. The model generalizes well to cross-domain corpora such as Wikipedia, arXiv, PubMed, and RST-DT, and can process long transcripts. Human evaluation further validates its reliability showing that 78% of its boundaries are rated correct ($\kappa = 0.72$). This proves the accuracy, efficiency, and adaptability of this method as a tool for segmenting lecture transcriptions, which improves summarization, indexing, and search capabilities on e-learning platforms.

**Keywords** Lecture Transcript Segmentation · Topic Segmentation · Neural Text Segmentation · Graph Neural Networks · Reinforcement Learning · Contrastive Learning

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## 1 Introduction

Text segmentation can generally be viewed as the process of breaking down large texts into smaller, coherent segments. Among them, Thematic Segmentation (TS) deals with the process of identifying where the theme changes. In this work, however, we address the more refined task of TS, which not only identifies topic boundaries but also captures the progression and continuity of themes in lecture delivery. TS, understood as dividing coherent text into coherent topic-based segments, has become an essential task for the comprehension of long-form content [1, 2]. Indeed, accurate segmentation enhances downstream applications like summarization, information retrieval, content indexing, and recommendation systems [3, 4]. While most research has focused on structured documents such as Wikipedia articles, news, and scientific papers, lecture video transcripts have unique challenges [5, 6]. Compared with written text, lecture transcripts contain disfluencies, repetitions, incomplete sentences, and colloquial expressions. Additionally, transitions in discourse topics are usually subtle and occur when a speaker moves from examples and/or explanations and/or concepts in a gentle manner without using a sudden transition phrase or pause. The subtleties of discourse and lecture transcripts make automatic discourse segmentation an even more challenging and essential task in e-learning systems such as MIT OpenCourseWare, VideoLectures.NET, Coursera, and TED-Ed.

Current approaches have three primary limitations (Fig. 1): First, the surface-word overlap approaches of LIWC/TxtTiling function entirely on the surface-word overlap level of text [7]. Although beneficial in identifying repeating vocabulary, these approaches do so by ignoring semantic cohesion that can be represented by synonyms and paraphrase. Second, transformer encoder architectures in BERT, like Longformer, maintain semantic understanding but require breaking transcripts down into fixed-size windows for processing

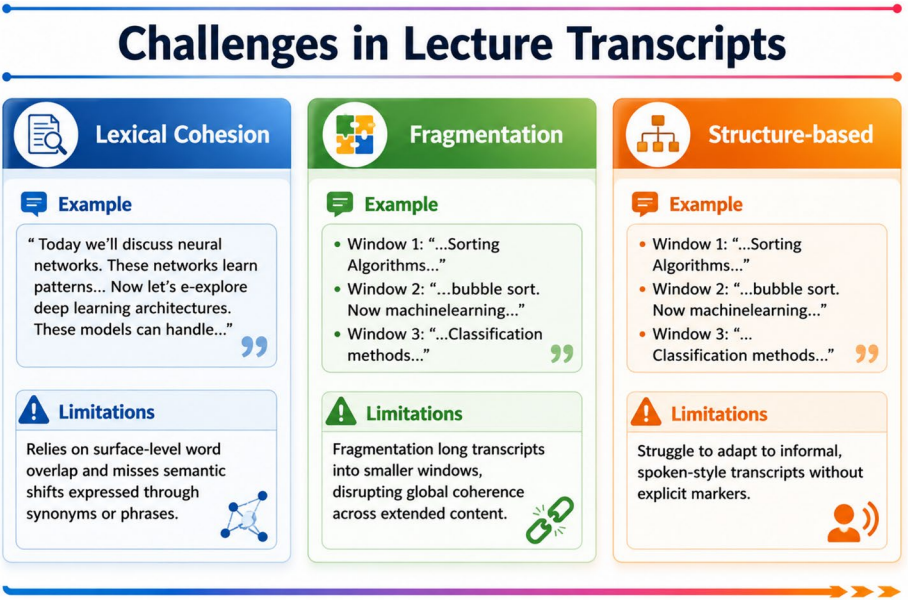



**Fig. 1** Limitations in Lecture Transcript Segmentation

lengthy transcripts [8, 9], even in their latest long-context versions [10]. Although windowing improves processing of lengthy transcripts, this defeats the purpose of maintaining coherence. The third limitation is that approaches designed for adaptation scenarios such as WikiSection perform well in organized texts but poorly in the unorganized, spoken format of lecture transcripts, which do not have heading or paragraph markers to indicate transitions.

To overcome these challenges, we propose this approach, a four-stage hybrid framework tailored for lecture transcripts. It includes:

**Long-context semantic encoding**: Using a modified SBERT encoder (all-mpnet-base-v2) fine-tuned on lecture data, combined with an overlapping-window strategy ($\omega = 2048$ tokens, 512-token overlap) to preserve semantic coherence across entire transcripts [10].
**Multi-scale neural segmentation with contrastive learning**: Capturing boundaries at sentence, paragraph, and topic levels, reflecting shifts from broad concepts to fine-grained examples [11, 12].
**Graph-based refinement**: Representing segments as nodes with edges weighted by semantic similarity, improving boundary consistency across distant segments [13, 14].
**Reinforcement learning with domain-aware rewards**: Optimizing predicted boundaries using metrics such as boundary consistency and semantic coherence [15].

Beyond methodological contributions, a key contribution of this work is the introduction of a Synthetic Lecture Transcript Benchmark comprising 210 lecture transcripts (around 1 million words, 1260 topic boundaries) mined from lectures on NPTEL, Coursera, and YouTube. This benchmark complements other publicly available benchmarks such as VideoLectures.NET, MIT-OCW, and TED-Ed, which are used not only as a training dataset but also for benchmarking. Contributions made in this work can be listed as follows:

**Integrated Hybrid Architecture:** Thematic Segmentation is presented to integrate long-context semantic embeddings, multi-level contrastive segmentation, coherence enhancement via graph, and learning by reinforcement into a task-specific approach to segment lecture scripts into themes.
**Lecture-Transcript-Oriented Design:** In contrast to previous methods that considered only semantic similarities, structural modeling, or boundary optimization separately, employs the above-mentioned complementary mechanisms in one framework for solving the problem.
**Dataset Contribution:** Synthetic Lecture Transcript Benchmark is proposed, which consists of 210 lecture transcripts annotated by humans from various educational fields.
**Empirical Validation:** We validate our approach with thorough experiments and ablations on the effectiveness of an integrated architecture that enhances performance.
**Broader applicability** – Proposing a framework that can easily accommodate various lecture data and be extended into other domains of long document segmentation.

Thematic content segmentation is not only helpful in improving the accuracy of content segmentation but also in facilitating adaptive learning and personalization of navigation through content and easy access to educational materials. The importance of education ecosystems supported by AI technologies, adaptive learning systems, and trustworthy AI

systems in learner engagement and accessibility has been stressed in numerous publications. Thematic content segmentation may also boost the efficiency of different educational applications, including automated summarization, recommendation systems, and learning analytics, thus facilitating accessibility of online education systems. Thematic content segmentation could be used for improvement of both educational content organization and education technology solutions. In this context, however, it must be noted that the use of components from proposed method, including SBERT, convolutional representations, graph neural networks, and reinforcement learning algorithms, is not something unheard-of in the machine learning literature. What is presented herein is not a totally new learning algorithm; rather, what makes this model special is the manner in which the aforementioned components are integrated into a single model for the solution of the theme segmentation problem with regard to lecture videos' transcripts. It is particularly advantageous because, as we have discussed earlier, each of the components fulfills its own purpose.

## 2 Related Work

Automated text segmentation has evolved from traditional methods of lexical cohesion to the latest models involving artificial neural networks, such as transformer models, to the largest language models [16]. Every new method of text segmentation has been developed to overcome the limitations of the previous method (Table 1).

### 2.1 Lexical Cohesion and Statistical Methods

Lexical cohesion algorithms, like TextTiling, are capable of segmenting the input data based on the repetition and co-occurrence of certain words, able to successfully identify the boundaries present in the structured input data [17, 18]. But the algorithms are inefficient in the expression of semantic changes present using synonyms or paraphrasing, which are common in lectures and are mostly missed [19]. DotPlot and C99 algorithms based on lexical similarity and rank-based clustering rely on the apparent word similarity and are inefficient for informal lectures as the speech might be disfluent and the vocabulary might be varying [20]. Lexical chain approaches try to follow words related in meaning through a document to identify topic boundaries. However, chains may reach across long transcripts, which introduces boundary ambiguity and can lower segmentation accuracy in extended lecture content [21]. The statistical or probability-based approaches, which involve Bayesian segmentation and Hidden Markov Models (HMM), attempt to predict boundary points based on the probabilities of word pairs or underlying topic patterns [22, 23]. While these approaches may seem more theoretically consistent than word co-occurrences, they have been found to make heavy assumptions that cannot be easily satisfied and fail to scale to long lectures due to varying topic lengths and implicit transition points.

### 2.2 Supervised Learning and Artificial Neural Networks

Classifier-based segmentation and Neural CPS use lexical and embedding features to model boundary predictions [24, 25]. While this increases performance on segmented text data, it does not generalize to lecture transcripts in a speech style because of differences in feature

**Table 1** Comparison of topic segmentation methods from 1997–2025 showing their approaches, datasets, strengths and limitations

| Method | Year | Approach | Dataset | Strengths | Limitations |
|---|---|---|---|---|---|
| TextTiling | 1997 | Lexical cohesion based on word repetition | Scientific papers | Fast, Simple | Surface-level, No semantics |
| C99 | 2000 | Rank-based clustering with similarity matrix | News articles | Good for structured text | Parameter sensitive, Lexical only |
| TopicTiling | 2012 | Sentence embeddings with semantic similarity | Wikipedia | Semantic aware, Embedding-based | Averaging loss, Short context |
| Neural CPS | 2018 | Neural classifier with lexical and embedding features | Mixed corpus | Supervised learning, Feature rich | Requires annotation, Domain specific |
| BERT-based | 2019 | Contextual embeddings with similarity scoring | WikiSection | Contextual understanding, Pre-trained | Windowing issues, Memory limits |
| WikiSection | 2020 | Semantic similarity with structural cues | Wikipedia | Structure aware, Multi-scale | Formal text only, Needs headings |
| Longformer | 2020 | Sparse attention for long sequences | Long documents | Long context, Efficient attention | Still fragments, Memory bound |
| RDA-WOA RNN+MHT | 2023 | Hybrid optimization-guided RNN with MHT for sequential feature extraction | Motion detection benchmarks | Robust temporal feature extraction; occlusion handling | Task-specific to video; high MHT cost |
| Fine-tuned AlexNet | 2023 | Transfer learning with fine-tuned CNN for anomaly detection | Crowd video datasets | Domain-adapted features; efficient transfer learning | Video-domain specific; limited architecture depth |
| LLM-guided Pop. RL | 2024 | LLM priors guide population-based RL for adaptive hyperparameter optimisation | Diverse optimisation benchmarks | Scalable; accelerates RL convergence | LLM inference cost; requires task-specific reward design |
| Privacy-Preserved Federated RS | 2024 | Federated DNN+TF-IDF for decentralised recommendation with data privacy | Flipkart smartphone reviews (13,000+) | 96.68% accuracy; user data stays local; privacy by design | Federated communication overhead; TF-IDF lexical only |

distributions and a lack of ability to model sentence structures and dependencies with a sufficient dataset [26]. Embeddings and transformer models have partially remedied the semantic shortcomings of traditional approaches [27]. With regard to TopicTiling, for instance, the semantic similarity between sentence embeddings is measured in terms of topic boundaries, which enhances the detection of paraphrased topics [28]. Nevertheless, the averaging process of embeddings in longer texts tends to limit the detection of topics within lectures. BERT models have traditionally relied on contextual embeddings based on a sentence or paragraph framework, which is segmented in terms of disjoint windows while transcribing lectures [29, 30].

Recently, significant progress has been made in feature extraction and representation learning methods, which have proven to be effective hybrid deep learning techniques across various application domains. For instance, a recurrent neural network based on the hybrid RDA-WOA method was designed for human motion recognition through multiple hypothesis tracking under occlusion conditions [31, 63], highlighting the importance of adaptive feature extraction and temporal dependency modeling. Fine-tuned AlexNet networks have also demonstrated effectiveness in detecting crowd anomalies using learned visual representations in complex video scenes [32, 64]. Furthermore, large language model-guided

population-based reinforcement learning has emerged as an adaptive optimization strategy for hyperparameter tuning [33, 65]. Although these studies address different application domains, they collectively emphasize the importance of robust feature extraction, representation learning, and adaptive learning mechanisms for improving model performance. These broader methodological insights are also relevant to transcript-based topic segmentation, where effective semantic representation, feature learning, contextual modeling, and adaptive decision-making play important roles in accurate topic boundary identification.

### 2.3 Transformer-based and Long-Context models

Despite using sparse attention models in order to increase the length of input sequences, memory constraints necessitate the use of windowing when processing extremely long speeches; this causes an inconsistent understanding process and inconsistencies in topic boundary detection [34, 35]. Although the SBERT cluster algorithms like k-means, DBSCAN, and hierarchical methods provide better boundaries semantically, they are still parameter-dependent and lack multiscale resolution capability along with sensitivity [36].

### 2.4 Domain Adaptation and Hybrid approaches

Both domain-specific and hybrid strategies aim to fill these gaps by adding structure and multiple phases of processing into the segmentation process [37]. For instance, WikiSection segments text based on semantics and structural information such as headings, and although it performs admirably on edited documents, it fails to perform adaptively on lectures without any structural markers [38]. Similarly, Graph and reinforcement-learning hybrids treat segments as nodes of a graph and optimize boundaries with respect to rewards for coherence [39, 40], but are usually tested on a very limited data set.

### 2.5 Large Language Models

Modern LLMs such as those based on GPT architectures are characterized by their ability to capture context-awareness and attention with semantic shift detection capabilities in longer contexts [41, 42]. Highly successful for most NLP applications, but computationally intensive for lecture transcription, they need sliding windows for even longer documents and do not target multi-scale boundaries between topics in education [43].

### 2.6 Data complexity and Privacy

In addition to segmentation quality, recent research has highlighted the importance of data complexity and privacy-aware intelligent systems. For example, data complexity analysis has been utilized to investigate the dependence of machine learning models on brain MRI datasets for brain tumor and Alzheimer's disease classification, demonstrating that dataset characteristics can significantly influence model behavior and predictive performance [44, 66]. Similarly, privacy-preserving decentralized recommendation systems have been proposed to provide accurate smartphone recommendations while protecting sensitive user information through distributed learning mechanisms [45, 67]. Although these studies

address application domains different from lecture transcript segmentation, they emphasize two broader challenges that are increasingly relevant across AI systems: robustness under varying data complexity and responsible handling of user-related information. In the context of thematic segmentation, lecture transcripts present their own complexity arising from long-context dependencies, semantic variability, gradual topic transitions, disfluencies, and transcription noise. While the proposed framework operates on publicly available educational transcripts and therefore does not directly involve sensitive personal data, future deployments within personalized e-learning environments may benefit from complexity-aware evaluation strategies and privacy-preserving learning paradigms.

### 2.7 Persistent limitations: Theoretical Motivation for the Integrated Framework

Despite these advances, all methods still share the following critical limitations: reliance on surface lexical features, fragmentation of long-context information, assumption of structured input, lack of annotated lecture datasets, and lack of sensitivity to multi-scale topic shifts. Current research efforts have explored each of these approaches separately in regard to various problems related to segmentation. To the best of our knowledge, however, there is currently no existing approach that combines all of the above approaches into a coherent solution for thematic segmentation of lectures from videos. In other words, what should be recognized as the novelty of this approach is the design of a novel architecture for the problem rather than any specific algorithmic advancements. In fact, the segmentation of lectures from videos includes three interrelated challenges—semantic understanding of the content given a long context; detecting the topic transitions between different temporal contexts, and ensuring global consistency of themes across the whole lecture transcript. To address all of these limitations, the proposed method here consists of four stages: each stage compensates for the weakness of others in segmenting a lecture transcript. Long-context semantic encoding considers semantic contextuality which cannot be simply detected by word similarities. The multi-scale contrastive learning method can find theme transitions across multiple scales while differentiating true topic transitions from local semantic variations. Graph-based refinement incorporates global reasoning to learn long-term semantic dependency, frequent concepts, and discourse structures that can only be expressed in terms of graph structure. Lastly, reinforcement learning conducts optimization in the whole sequence, evaluating each decision of boundary positions in the whole segmentation quality. In this way, these four stages form a complementary hierarchy where semantics are encoded first to capture the context, candidate themes are identified with contrastive learning, graph-based refinement guarantees theme consistency, and reinforcement learning optimizes the document-level segmentation quality.

Building on these observations, this work addresses these chronic limitations by integrating the strengths of four complementary methods into a single framework. Coherent semantic encoding over long contexts addresses fragmentation artifacts characteristic of windowed transformer and LLM-based approaches. Multi-scale neural segmentation with contrastive learning enables the detection of topic boundaries at sentence, paragraph, and topic levels of resolution, thereby allowing sensitivity to subtle and hierarchical transitions that classical and embedding-based methods usually fail to detect. Graph refinement models segments as a graph of nodes connected by edges that show similarity between segments. This pro-

vides global coherence by considering similarity between non-adjacent segments. Another improvement is the application of reinforcement learning with domain-related reward functions such as boundary consistency and coherence. This directly optimizes boundary placement using intrinsic segmentation-quality rewards such as semantic coherence, semantic progression, redundancy reduction, and segment-length regularization to compensate for the lack of available training data for lectures that have an informal structure. By harnessing these approaches together, this work is able to leverage the best-of-breed capabilities of semantic embeddings, multi-scale modeling, graph reasoning, and adaptive optimization and thus offers an efficient and accurate method for segmenting lecture transcripts across various educational domains.

## 3 Methodology

Thematic Segmentation is designed as a four-stage hybrid framework that reconceptualizes segmentation as thematic progression analysis rather than simple boundary detection. Unlike generic approaches, captures gradual topic shifts, scaffolded explanations, and interconnected concept development typical of lecture delivery. The framework integrates long-context encoding, multi-scale segmentation, graph-based refinement, and reinforcement learning into a unified pipeline (Fig. 2), enabling accurate, efficient, and meaningful segmentation of lecture transcripts.

### 3.1 Long-Context Semantic Encoding

All transcripts are preprocessed before model training and evaluation. The text is converted to lowercase, Unicode-normalized, and stripped of redundant whitespace [46]. Sentences are segmented using a rule-based tokenizer to ensure consistent sentence indices across all experiments [26]. Speaker tokens and timestamp tokens are removed, while slide-change

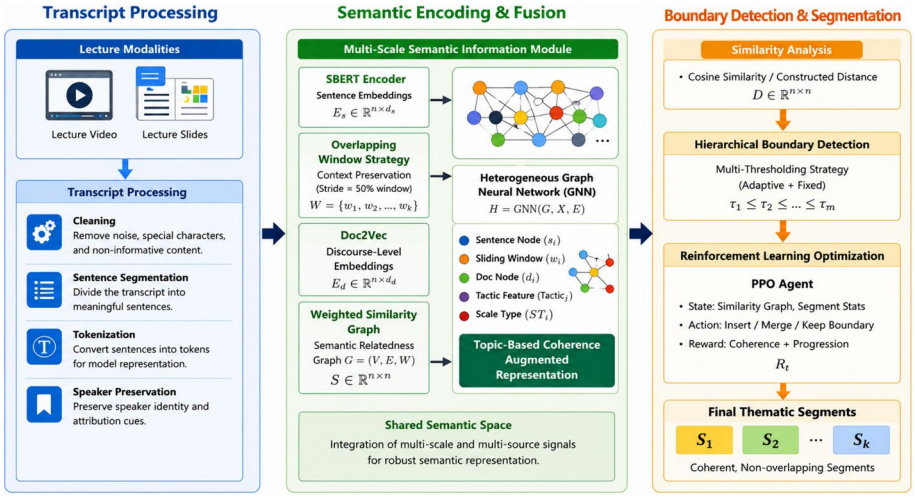



**Fig. 2** Multi-Stage Deep Learning for Topic Boundary Optimization

tokens, where present, are retained as structural markers. Content words are used to compute lexical cohesion.

To handle transcripts exceeding standard model input capacity, an overlapping-window strategy (Eq. (1) and Algorithm 1) was employed with a maximum window size of $\omega = 2048$ tokens and a sliding step of $\lambda = 1536$ tokens, creating a 512-token overlap to preserve semantic continuity across boundaries [47]:

$$W_i = \{s_i, s_{i+1}, \ldots, s_{i+\omega-1}\} \tag{1}$$

A modified Sentence-BERT (SBERT) encoder, fine-tuned, was used to generate semantically meaningful sentence embeddings while addressing computational limitations in long-context processing (Eq. (2)) [48]. For each sentence $s_i$, token embeddings were extracted, and mean pooling was applied to compute sentence-level embeddings:

$$u = \frac{1}{m} \sum_{i=1}^{m} h_i \tag{2}$$

where $\bar{h}_i \in \mathbb{R}^{768}$ denotes token embeddings, m is the number of tokens in a sentence, and $\bar{u} \in \mathbb{R}^{768}$ is the resulting sentence embedding. Next, classification was performed to identify functions beyond surface-level semantics.

### 3.1.1 Pedagogical Marker Classification

Marker classification was formulated as a six-class supervised learning problem consisting of INTRODUCTION, DEFINITION, EXAMPLE, TRANSITION, SUMMARY, and CONCLUSION discourse functions. Training labels were derived from manually annotated lecture transcripts. Sentence embeddings generated by the long-context encoder served as input features to a multinomial logistic regression classifier. The classifier was optimized using cross-entropy loss and trained using the Adam optimizer. Predicted marker labels were subsequently incorporated into graph construction and boundary refinement stages. A multinomial logistic regression classifier was employed to identify pedagogical discourse markers such as INTRODUCTION, DEFINITION, EXAMPLE, TRANSITION, SUMMARY, and CONCLUSION. Marker predictions were generated using lexical and semantic discourse cues extracted from lecture transcripts and were incorporated as auxiliary features for subsequent segmentation stages(Eq. (3)):

$$\text{markertype} = \text{softmax}(W_m u + b_m) \tag{3}$$

where marker_type $\in$ {INTRODUCTION, DEFINITION, EXAMPLE, TRANSITION, SUMMARY, CONCLUSION}, $W_m \in \mathbb{R}^{6\times768}$ are learned weights, and $b_m \in \mathbb{R}6$ are bias terms. Marker labels were generated using weakly supervised discourse-cue patterns commonly observed in lecture transcripts rather than a separate manually annotated dataset. Representative cue expressions associated with INTRODUCTION, DEFINITION, EXAMPLE, TRANSITION, SUMMARY, and CONCLUSION were used to generate the initial

labels employed by the marker classifier. The marker categories serve as auxiliary discourse indicators rather than primary supervision signals for segmentation. Marker predictions are used to enrich sentence representations and provide pedagogical context for subsequent segmentation stages. Their contribution is complementary to semantic embeddings, graph refinement, and reinforcement learning optimization. The marker classifier identifies pedagogical discourse functions including INTRODUCTION, DEFINITION, EXAMPLE, TRANSITION, SUMMARY, and CONCLUSION. Marker predictions are incorporated as auxiliary discourse features and provide additional contextual information regarding lecture structure. These features support subsequent segmentation stages but do not directly determine topic boundaries. Similarity computations were then applied to construct a sentence similarity matrix (Eq. (4)). Sentence embeddings were normalized to unit vectors, and cosine similarity was computed for all sentence pairs [49]:

$$\mathrm{Sim}(i,j) = \frac{v_i \cdot v_j}{\|v_i\|\|v_j\|} \tag{4}$$

where $\bar{v}_i = \frac{\bar{u}_i}{\|\bar{u}_i\|_2}$ is the L2-normalized embedding of sentence si. The resulting similarity matrix $S \in \mathbb{R}^{n \times n}$ encodes pairwise relationships between all sentences. Finally, a weighted adjustment was applied to incorporate positional proximity and lexical cohesion cues (Eq. (5)). Jaccard similarity of content words was integrated with a positional decay factor to refine similarity scores:

$$Sim_{weighted}(i,j) = Sim(i,j) \cdot \exp\left(-\frac{|i-j|}{50}\right) \cdot (1 + 0.3 \cdot \mathrm{overlap}_{score}(s_i, s_j)) \tag{5}$$

This ensures that semantically similar but distant sentences are downweighted, while sentences with lexical overlap receive boosted similarity. Sample outputs are described in Table 2.

Input: Transcript T = {$s_1$, $s_2$, ..., $s_n$}
Output: Sentence embeddings V = {$\bar{v}_1$, $\bar{v}_2$, ..., $\bar{v}_n$}, Similarity matrix S_weighted

1.W = partition_transcript (T, ω=2048, λ=1536) // Overlapping windows
2.U = []
3.for each window $W_i$ in W do
4. U.extend ([mean_pool(SBERT($s_j$)) for $s_j$ in $W_i$]) // Sentence embeddings
5.M = [argmax (softmax($W_m \times \bar{u}_i + b_m$)) for $\bar{u}_i$ in U] // Sentence classification
6.V = [$\bar{u}_i$ / $\|\bar{u}_i\|_2$ for $\bar{u}_i$ in U] // L2 normalization
7.S = [[dot ($\bar{v}_i$, $\bar{v}_j$) for $\bar{v}_j$ in V] for $\bar{v}_i$ in V] // Base similarity matrix
8.for i in range(n):
9. for j in range(n):
10. overlap = jaccard_similarity (content_words($s_i$), content_words ($s_j$))
11. S_weighted[i,j] = S[i,j] × exp(-|i-j|/50) × (1 + 0.3 × overlap)
12. return V, S_weighted
13.End

**Table 2** Sample output of modified SBERT for transcript processing

| Component | Output |
|---|---|
| Sentence Embeddings | $s_1$: [0.23, −0.15, 0.41, …] ∈ $\mathbb{R}^{768}$ $s_2$: [0.31, −0.09, 0.38, …] ∈ $\mathbb{R}^{768}$ $s_3$: [0.19, −0.22, 0.35, …] ∈ $\mathbb{R}^{768}$ |
| Markers | $s_1$: INTRODUCTION (0.85) $s_2$: DEFINITION (0.91) $s_3$: TRANSITION (0.78) |
| Similarity Matrix | $s_1$–$s_2$: 0.84, $s_1$–$s_3$: 0.62 $s_2$–$s_3$: 0.71 |
| Semantic Memory | [machine_learning, AI, supervised, algorithms, discussion] |

### 3.2 Multi-Scale Neural Topic Segmentation with Contrastive Learning

Hierarchical boundary detection is achieved at this level through a multi-scale CNN that employs contrastive learning tasks for discerning authentic topic transitions from noise [50]. This task involves processing the embedding representation of each sentence created by Stage 1 at various time scales, allowing for the detection of both local and global thematic boundaries associated with long form transcripts (Algorithm 2). The multiscale model examines boundaries in parallel across multiple temporal scales. The detection process uses sliding windows of various lengths to detect local topic changes, intermediate changes in concepts, and global theme organization (Eq. (6)). Multiscale feature extraction performs convolution over sentence embeddings on different temporal scales [51]. For input embeddings $V = \{\bar{v}_1, \bar{v}_2, \ldots, \bar{v}_n\}$ from Stage 1, the system applies parallel convolution operations to capture temporal patterns at multiple resolutions:

$$F_k = \text{Conv1D}_k(V) = \sigma(W_k * V + b_k) \tag{6}$$

where $k \in \{3, 5, 7, 9\}$ represents kernel sizes capturing different temporal scales, $W_k \in \mathbb{R}^{dxk}$ are learnable convolution weights, $b_k \in \mathbb{R}^d$ are bias terms, and σ is the ReLU activation function.

#### 3.2.1 Contrastive Pair Sampling Strategy

Contrastive learning trains the model to distinguish true thematic boundaries from false positives using positive and negative boundary examples [52]. The contrastive loss encourages similar representations for genuine boundaries while pushing apart representations of non-boundaries(Eq. (7)):

$$L_{\text{contrastive}} = -\log\left(\frac{\exp\left(\text{sim}(\mathbf{f}_{\text{anchor}}, \mathbf{f}_{\text{positive}})/\tau\right)}{\sum_j \exp\left(\text{sim}(\mathbf{f}_{\text{anchor}}, \mathbf{f}_{\text{negative},j})/\tau\right)}\right) \tag{7}$$

where $\boldsymbol{f}_{anchor}$ denotes the feature representation of a candidate boundary position, $\boldsymbol{f}_{positive}$ represents the feature vector corresponding to a gold-standard topic boundary in the training data, and $\boldsymbol{f}_{negative}$ denotes the feature vector of a non-boundary position sampled from the same transcript. The contrastive loss minimizes the distance between $\boldsymbol{f}_{anchor}$ and $\boldsymbol{f}_{positive}$ while maximizing the distance between $\boldsymbol{f}_{anchor}$ and $\boldsymbol{f}_{negative}$, thereby learning discriminative

representations for thematic boundary detection. During training, one positive sample and five negative samples were selected for each anchor position. Positive samples correspond to gold-standard topic boundaries, while negative samples are randomly drawn from non-boundary positions within the same transcript. This strategy encourages robust separation between boundary and non-boundary representations.

Attention-based feature fusion [29] combines multi-scale features to produce unified boundary representations (Eq. (8a), (8b)). The attention mechanism weighs different scales based on their relevance for boundary detection:

$$\alpha_k = \frac{\exp\left(W_{\text{att}}^{\text{T}} \tanh\left(W_k F_k + b_{\text{att}}\right)\right)}{\sum_j \exp\left(W_{\text{att}}^{\text{T}} \tanh\left(W_j F_j + b_{\text{att}}\right)\right)} \tag{8a}$$

$$F_{\text{fused}} = \sum_k \alpha_k \cdot F_k \tag{8b}$$

where $W_{att} \in \mathbb{R}^{dxd}$, $b_{att} \in \mathbb{R}^{d}$ are attention parameters, and F_fused $\in \mathbb{R}^{nxd}$ represents the unified multi-scale features. Auxiliary feature integration combines neural features with marker and similarity features from Stage 1 to refine boundary predictions(Eq. (9)) [53]:

$$F_{\text{enhanced}} = F_{\text{fused}} + \lambda_{\text{aux}} \cdot W_{\text{aux}} \cdot [f_{\text{maker}}; f_{\text{sim}}] \tag{9}$$

where $\lambda_{\text{aux}} = 0.3$ weights the auxiliary contribution, $W_{\text{aux}} \in \mathbb{R}^{dx(\text{marker_dim+sim_dim})}$ projects auxiliary features, and [;] denotes concatenation. Boundary classification [30] predicts thematic boundaries using the enhanced features through binary classification (Eq. (10)). The final boundary probability for position i is:

$$P(\text{boundary} \mid i) = \sigma(W_{\text{cls}} \cdot F_{\text{enhanced}}[i] + b_{\text{cls}}) \tag{10}$$

where $W_{\text{cls}} \in \mathbb{R}^{d}$, $b_{\text{cls}} \in \mathbb{R}$ are classification parameters, and positions with P(boundary|i) > 0.5 are identified as thematic boundaries (sample output in Table 3).

Input: Sentence embeddings V = {$\vec{v}_1$, $\vec{v}_2$, ..., $\vec{v}_n$}, Marker_features M, Weighted similarity matrix S_weighted
Output: Boundary probabilities P(boundary|i) for all

1.For each kernel size k ∈ {3,5,7,9} do
2. F_k = Conv1D_k(V) // Multi-scale convolution
3.end for
4.Compute attention weights α_k across scales
5.F_fused = Σ_k α_k · F_k // Attention-based feature fusion
6.F_enhanced = F_fused + λ_aux · W_aux · [M; S_weighted]
// Integrate auxiliary features
7.Train using contrastive objective:
8. Minimize L_contrastive(f_anchor, f_positive, f_negative)
9.For each sentence i in transcript do
10. P(boundary|i) = sigmoid(W_cls · F_enhanced[i] + b_cls)
11.end for
12.return P(boundary|i) for all i
13.End

**Table 3** Multi-scale Neural Topic Segmentation output

| Component | Output |
|---|---|
| Multi-scale Features | $F_3$: [0.45, 0.23, −0.12, …] $\in \mathbb{R}^{256}$, $F_5$: [0.31, −0.08, 0.67, …] $\in \mathbb{R}^{256}$, $F_7$: [0.19, 0.44, −0.33, …] $\in \mathbb{R}^{256}$, $F_9$: [0.52, −0.17, 0.28, …] $\in \mathbb{R}^{256}$ |
| Attention Weights (α) | $\alpha_3 = 0.15$, $\alpha_5 = 0.35$, $\alpha_7 = 0.28$, $\alpha_9 = 0.22$ ($\Sigma\alpha = 1.0$) |
| Fused Features | F_fused = 0.15·F3 + 0.35·F5 + 0.28·F7 + 0.22·F9 → [0.38, 0.11, 0.05, …] $\in \mathbb{R}^{256}$ |
| Boundary Probabilities | $s_1 \rightarrow s_2$: 0.12 (Non-boundary) $s_2 \rightarrow s_3$: 0.78 (Boundary) $s_3 \rightarrow s_4$: 0.23 (Non-boundary) $s_4 \rightarrow s_5$: 0.85 (Boundary) |
| Detected Boundaries | [2, 4] → Boundaries after $s_2$ and $s_4$ (e.g., "Deep learning → CNNs", "CNNs → RNNs") |

### 3.3 Graph Neural Network-based Refinement

The third stage constructs a Heterogeneous Graph representation to model complex semantic relationships and coherence patterns that go beyond linear sentence sequences [35, 54]. The approach refines boundary predictions from previous stage by incorporating conceptual dependencies, cross-reference relationships, and thematic callbacks through a graph attention network (GAT)-based processing [55]. The graph construction process creates a heterogeneous network where nodes represent sentences, concepts, and marker features, while edges capture semantic similarity, structural relations, and temporal dependencies [36, 56]. This representation enables modeling of non-linear transcript patterns, such as forward references and thematic recurrence (Algorithm 3).

Graph node initialization creates embeddings for different entity types (Eq. (11)). Sentence nodes use enhanced embeddings from Stage 1, concept nodes represent extracted key terms, and marker nodes encode classification features.

$$N_{\text{sent}}[i] = \bar{v}_i \in \mathbb{R}^{768} \tag{11}$$

where $\bar{v}_i$ are sentence embeddings from Stage 1, $\text{concept}_j$ are extracted key terms, and $\text{marker}_k$ are functional marker features.

#### 3.3.1 Concept Node Extraction

Concept nodes represent salient thematic concepts occurring in the transcript. Candidate concepts are extracted using noun-phrase chunking and part-of-speech tagging. The extracted noun phrases are ranked using TF-IDF scores, and the top-K concepts (K = 15) are retained as concept nodes. Examples include domain-specific concepts such as *machine learning*, *neural network*, and *gradient descent*. These concept nodes provide semantic anchors that connect related sentences discussing the same concept. Edge construction establishes weighted connections based on semantic similarity, temporal proximity, and auxiliary relations (Eq. (12a)):

$$w(i,j) = \alpha_{\text{sem}} \cdot sim(N_i, N_j) + \alpha_{\text{temp}} \cdot \exp\left(-\frac{|i-j|}{\sigma_{\text{temp}}}\right) + \alpha_{\text{aux}} \cdot \text{relation}(i,j) \tag{12a}$$

where $\alpha_{\text{sem}}$=0.5, =0.3, $\alpha_{\text{aux}}$=0.2, and $\sigma_{\text{temp}}$=5 controls temporal decay. The auxiliary relation term captures semantic associations that are not represented by embedding similarity alone. Specifically(Eq. (12b)),

$$\text{relation}(i,j) = \begin{cases} 1, & \text{if a sentence contains a concept node or shares a discourse-marker association} \\ 0, & \text{otherwise} \end{cases} \tag{12b}$$

A sentence node is connected to a concept node when the corresponding concept appears within the sentence. Similarly, sentence nodes are linked to marker nodes according to the discourse function predicted in Stage 1 (e.g., INTRODUCTION, DEFINITION, TRANSITION, SUMMARY). These auxiliary relations enrich the graph with semantic and discourse-level information beyond pairwise sentence similarity. A simple sentence-similarity graph models only pairwise semantic similarity between sentences. However, lecture transcripts frequently contain recurring concepts, discourse markers, and long-range thematic dependencies that cannot be fully represented through sentence similarity alone. To overcome this problem, uses a heterogeneous graph with three types of nodes, namely sentence nodes, concept nodes, and marker nodes. Concept nodes serve as semantic reference points for sentences about the same theme. On the other hand, marker nodes carry discourse information representing discourse functions for thematic shifts. With a more complex structure, the model allows information flow on three levels: semantic, structural, and discourse levels, making boundary refinement better than only based on the sentence graph representation. Graph generation is composed of three processes, including (1) initialization of sentence nodes via SBERT embeddings, (2) generation of concept nodes through noun-phrase chunking and TF-IDF ranking, and (3) edge generation by semantic, temporal, and auxiliary semantic similarities. Graph attention message passing (GAT): Each node aggregates messages from neighbors, with attention weighting neighbor importance (Eq. (13a)) [57]:

$$m_{ij}^{(l)} = \alpha_{ij}^{(l)} \cdot W_{\text{msg}}^{(l)} N_j^{(l)} \tag{13a}$$

where $W^{(l)}_{msg} \in \mathbb{R}^{dxd}$ are learnable weights, and $\alpha^{(l)}_{ij}$ are normalized attention coefficients (Eq. (13b)):

$$\alpha_{ij}^{(l)} = \frac{\exp\left(\text{LeakyReLU}\left(\mathbf{a}^\top \left[W_{\text{msg}}^{(l)} N_i^{(l)} \parallel W_{\text{msg}}^{(l)} N_j^{(l)}\right]\right)\right)}{\sum_{k \in \text{neighbors}(i)} \exp\left(\text{LeakyReLU}\left(\mathbf{a}^\top \left[W_{\text{msg}}^{(l)} N_i^{(l)} \parallel W_{\text{msg}}^{(l)} N_k^{(l)}\right]\right)\right)} \tag{13b}$$

Node update (Eq. (14)):

$$N_i^{(l+1)} = \sigma\left(N_i^{(l)} + \sum_{j \in \text{neighbors}(i)} m_{ij}^{(l)}\right) \tag{14}$$

where σ is ReLU activation, and residual connections preserve original information.

Boundary refinement (Eq. (15)): Stage 2 probabilities are adjusted using graph-derived coherence:

$$P_{\text{refined}}(\text{boundary} \mid i) = \beta \cdot P_{\text{stage2}}(\text{boundary} \mid i) + (1 - \beta) \cdot \text{coherence_score} \quad (15)$$

where β=0.7 weights Stage 2 predictions and coherence_score measures graph-based consistency [58]. Sample output is mentioned in Table 4.

Input: Sentence embeddings V = {$\bar{v}_1$, $\bar{v}_2$, ..., $\bar{v}_n$},Stage 2 probabilities P_stage2, Concept nodes C, Marker features M
Output: Refined boundary probabilities P_refined(boundary|i)

1.Construct heterogeneous graph G = (N, E)
// Nodes = {sentences, concepts, markers}
// Edges = semantic similarity, temporal links, auxiliary relations
2.Initialize node embeddings:
3. N_sent[i] = $\bar{v}_i$ // Sentence embeddings from Stage 1
4. N_concept[j] = concept_j // Key terms extracted from transcript
5. N_marker[k] = marker_k // Marker features from Stage 1 classification
6.For each edge (i,j):
7. w(i,j) = α_sem·sim(N_i, N_j)
+ α_temp·exp(-|i-j|/σ_temp)
+ α_aux·relation(i,j)
// Edge weights combine semantic, temporal, and auxiliary relations
8.For l = 1 to L do // Multi-layer GAT propagation
9. For each node i:
10. For each neighbor j:
11. Compute attention coefficient α_ij^(l) // Normalized attention score for edge (i,j)
12. Message m_ij^(l) = α_ij^(l) · W_msg^(l) N_j^(l)
// Weighted neighbor embedding after transformation
13. Update node embedding:
14. N_i^(l+1) = σ(N_i^(l) + Σ_j m_ij^(l)) // Residual connection + ReLU activation
15.Compute coherence_score(i) from final graph embeddings
//Evaluates boundary likelihood based on graph structure
16.Refine probabilities:
17. P_refined(boundary|i) = β·P_stage2(boundary|i)
+ (1-β)·coherence_score(i) // Combine Stage 2 prediction with graph-based refinement
18. return P_refined(boundary|i) for all i
19.End

### 3.4 Reinforcement Learning Optimization

Reinforcement learning is employed as a sequence-level optimization stage rather than a primary boundary detector. Stages 1–3 generate candidate boundaries using semantic, structural, and graph-based evidence, while PPO refines these predictions by optimizing global segmentation quality. Unlike local classifiers that evaluate boundaries independently, the RL agent considers the cumulative effect of boundary decisions on semantic coherence, progression, redundancy, and segment balance across the entire transcript. Consequently, RL functions as a document-level optimization mechanism that complements rather than replaces supervised boundary detection. Reinforcement learning is used for this stage for the optimal segmentation by means of reward functions which correspond to segmentation-

**Table 4** Sample output of graph-based refinement using (GAT)

| Component | Output |
|---|---|
| Sentence Nodes | $N_1$: [0.21, −0.14, 0.38, …]$\in\mathbb{R}^{768}$ $N_2$: [0.29, −0.08, 0.41, …]$\in\mathbb{R}^{768}$ |
| Concept Nodes | {machine_learning, CNN, RNN} |
| Marker Nodes | {INTRODUCTION, DEFINITION, TRANSITION} |
| Edge Weights | w(1,2) = 0.64, w(2,3) = 0.71, w(3,4) = 0.58 |
| GAT Attention Coefficients ($\alpha_{ij}$) | $\alpha_{12}$ = 0.32, $\alpha_{23}$ = 0.41, $\alpha_{34}$ = 0.27 ($\Sigma\alpha$ = 1.0 for each node) |
| Message Passing | $m_{23} = \alpha_{23} \cdot W_msg\ N_3$ = [0.12, 0.08, −0.04, …] |
| Updated Node Embeddings | $N_2'$ = [0.25, −0.05, 0.44, …] $N_3'$ = [0.28, −0.11, 0.40, …] |
| Stage 2 Probabilities | $s_2 \rightarrow s_3$: 0.78 (Boundary), $s_4 \rightarrow s_5$: 0.85 (Boundary) |
| Coherence Scores | $s_2 \rightarrow s_3$: 0.74, $s_4 \rightarrow s_5$: 0.81 |
| Refined Probabilities | $s_2 \rightarrow s_3$: 0.76, $s_4 \rightarrow s_5$: 0.83 |
| Final Boundaries | [2,4] → Boundaries after $s_2$ and $s_4$ (confirmed) |

quality criteria. The technique considers the segmentation process as a sequence decision-making task wherein the agent tries to place boundaries by maximizing segmentation quality via interaction with the transcripts (e.g., coherence and progress) [59] (Sample output was described in Table 5).

**Table 5** Sample output of Segmentation Optimization

| Component | Sample Output (example) |
|---|---|
| State ($S_3$) | [$\bar{v}_3$; *marker*$_3$; *graph_features*$_{i3}$; {NO,NO}]$\in\mathbb{R}$^*d*_state |
| Policy probs | $\pi(a\|s_3)$ = [0.15, 0.25, 0.60] |
| Selected action ($a_3$) | STRONG_BOUNDARY (sampled/greedy) |
| Reward components | R_coh = 0.62,<br>R_prog = 0.48,<br>R_red = −0.05,<br>R_len = 0.10 |
| Weights used | W_coh = 0.4,<br>W_prog = 0.3,<br>W_red = 0.2,<br>W_len = 0.1 |
| Total reward ($r_3$) | 0.4 × 0.62 + 0.3 × 0.48 + 0.2 × (−0.05) + 0.1 × 0.10 = 0.392 |
| Updated segmentation | boundary inserted after sentence 3; *segment_history*$_t$ updated |
| Final segmentation | Example final boundaries: indices [2, 3, 5] (after post-processing) |
| Policy diagnostics | avg. entropy = 0.92; avg. action-prob for STRONG = 0.34 (over episode) |
| Training metrics | PPO loss = −0.013, value loss = 0.09, avg. episode reward = 12.4 |
| Reward traces | per-step reward sequence: [0.02, 0.01, 0.44, 0.05, 0.36, …] |

### 3.4.1 Markov Decision Process(MDP) Formulation

The model uses a formulation where boundary placement is represented as a Markov Decision Process (MDP) in which states describe the current segmentation context, actions include boundary placements, and reward functions depend on segmentation quality [37, 60]. In this way, it is possible to train a model that takes into account the long-term effect of the segmentation and not only local similarities. State representation includes the current segmentation context as well as sentence, markers/classification, graph structure and partially completed segmentation data (Eq. (16)). The state vector includes:

$$s_t = [\bar{v}_t\,;\,\text{marker}_t\,;\,\text{graph_features}_t\,;\,\text{segment_history}_t] \in \mathbb{R}^{d_{\text{state}}} \tag{16}$$

where $\bar{v}_t$ is the current sentence embedding, *marker*$_t$ encodes classifier/marker information, *graph_features*$_t$ captures local graph structure, and *segment_history*$_t$ represents recent segmentation decisions. Action space definition specifies boundary placement decisions available at each position. The agent chooses from three discrete actions representing different boundary decisions based on segmentation confidence and thematic strength (Eq. (17)):

$$a_t \in \{\text{NO_BOUNDARY}, \text{WEAK_BOUNDARY}, \text{STRONG_BOUNDARY}\} \tag{17}$$

where NO_BOUNDARY indicates continuation of the current segment, WEAK_BOUNDARY suggests a tentative transition (may be conditioned on post-processing rules), and STRONG_BOUNDARY indicates a definitive thematic shift.

### 3.4.2 Reward Computation Strategy

Composite quality reward computation evaluates segmentation decisions using intrinsic segmentation-quality signals derived from the predicted segmentation itself rather than from gold-standard boundaries. Specifically, (Eq. (18a)). The instantaneous reward for ($s_t$, $a_t$) is (Eq. 18(b-e))

$$R_{\text{quality}}(s_t, a_t) = w_{\text{coh}} \cdot R_{\text{coh}} + w_{\text{prog}} \cdot R_{\text{prog}} + w_{\text{red}} \cdot R_{\text{redundancy}} + w_{\text{len}} \cdot R_{\text{length}} \tag{18a}$$

$$R_{\text{coh}} = \frac{1}{|S|} \sum_{s_i, s_j \in S} \cos(v_i, v_j) \tag{18b}$$

where **S** denotes the current predicted segment, $\mathbf{s_i}$ and $\mathbf{s_j}$ denote the i-th and j-th sentences belonging to segment **S**, respectively. Their corresponding SBERT embedding vectors are represented by $\mathbf{v_i}$ and $\mathbf{v_j}$. The cosine similarity between these embedding vectors quantifies the semantic relatedness of sentence pairs, thereby encouraging semantic coherence within the predicted segment.

$$R_{\text{prog}} = 1 - \frac{1}{K-1} \sum_{k=1}^{K-1} \cos(c_k, c_{k+1}) \tag{18c}$$

where **K** denotes the total number of segments in the predicted segmentation, $\mathbf{c_k}$ represents the centroid embedding of the k-th segment, and $\mathbf{c_{k+1}}$ represents the centroid embedding of the subsequent segment. The cosine similarity between $\mathbf{c_k}$ and $\mathbf{c_{k+1}}$ measures the semantic similarity of adjacent segments. Subtracting the average similarity from 1 rewards semantic progression by encouraging neighboring segments to be semantically distinct while maintaining topic continuity.

$$R_{\text{redundancy}} = -\frac{1}{K-1}\sum_{k=1}^{K-1}\cos(c_k, c_{k+1}) \tag{18d}$$

where **K** denotes the total number of predicted segments, $\mathbf{c_k}$ represents the centroid embedding of the k-th segment, and $\mathbf{c_{k+1}}$ represents the centroid embedding of the adjacent segment. The cosine similarity between $\mathbf{c_k}$ and $\mathbf{c_{k+1}}$ quantifies the semantic overlap between neighboring segments. The negative sign penalizes excessive similarity between adjacent segments, thereby reducing redundancy and encouraging each segment to convey distinct semantic content.

$$R_{\text{length}} = -\left(\frac{|L-\mu|}{\mu}\right) \tag{18e}$$

where **L** denotes the number of sentences in the current predicted segment and **μ** is the average segment length computed from the training data. This reward penalizes deviations from the expected segment length, promoting balanced segmentation. Importantly, these reward components are computed automatically from transcript embeddings and predicted boundaries without access to reference segmentations. Gold-standard boundaries are used only during supervised training of Stage 2 and for evaluation. Consequently, the reinforcement learning stage optimizes intrinsic segmentation quality and should be viewed as a post-processing optimization mechanism rather than a surrogate form of supervised learning. where $w_{\text{coh}} = 0.4, \quad w_{\text{prog}} = 0.3, \quad w_{\text{red}} = 0.2, \quad w_{\text{len}} = 0.1$ weight differentqualityaspects.Eachcomponent (e.g., $R_{\text{coh}}$: intra-segment cohesion; $R_{\text{prog}}$: topical progression; $R_{\text{red}}$: redundancy penalty; $R_{\text{len}}$: length regularizer) should be normalized and clipped before combination.

### 3.4.3 Proximal Policy Optimization(PPO)

Policy network training employs Proximal Policy Optimization (PPO) to learn optimal boundary placement strategies [61] (Eq. (19)). The policy network πθ(a|s) maps states to action probabilities using neural network parameters θ:

$$\pi_\theta(a_t \mid s_t) = \text{softmax}\left(W_{\text{poli}}\phi(s_t) + b_{\text{textpoli}}\right) \tag{19}$$

where$\phi s$ $\phi(s_t)$processes state features through hidden layers, $W_{\text{poli}} \in \mathbb{R}^{3\times d_{\text{hidden}}}$ and $b_{\text{poli}} \in \mathbb{R}^3$ project to the action logits. Policy optimization updates using the PPO loss combine policy improvement with value-function learning and entropy regularization [62]. The optimization balances exploitation of current knowledge with exploration of new strategies (Eq. (20))

$$L_{\mathrm{PPO}}(\theta) = \mathbb{E}_t \left[ \min \left( r_t(\theta)\hat{A}_t,\ \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t \right) \right] - c_{\mathrm{value}} L_{\mathrm{value}} + c_{\mathrm{entropy}} H(\pi_\theta) \quad (20)$$

where $r_t(\theta)$ is the probability ratio, $\hat{A}_t$ is the advantage estimate (use GAE), $\epsilon = 0.2$ is the clipping parameter, and $c_{\mathrm{value}} = 0.5$, $c_\mathrm{entropy} = 0.01$ weight auxiliary losses. (Note: the entropy term is subtracted here because the expression is a loss to be minimized; implementations that maximize an objective will sign it accordingly.) At each sentence position, the PPO agent selects either Boundary or Non-Boundary as its action. The reward function combines four components: semantic coherence ($w_{\mathrm{coh}} = 0.4$), topic progression consistency ($w_{\mathrm{prog}} = 0.3$), redundancy reduction ($w_{\mathrm{red}} = 0.2$), and segment length regularization ($w_{\mathrm{len}} = 0.1$). Semantic coherence is computed as the average cosine similarity among sentences within the predicted segment. Topic progression rewards boundaries that align with gold-standard thematic transitions. Redundancy penalties discourage unnecessary boundary insertions, while length regularization prevents excessively short or excessively long segments. The final reward is computed as the weighted sum of these four components.

Input: Sentence embeddings V = {$\bar{v}1$, $\bar{v}2$, ..., $\bar{v}n$},
Refined probabilities P_refined (Stage 3),
Graph features graph_features, Marker features M
Output: Final segmentation labels (boundary indices), trained policy $\pi$_θ, reward traces

// init networks and buffer
1.initialize π_θ, V_φ, buffer
// collect on-policy rollouts
2.for each transcript do
3.reset *segment_history*$_t$
4.for t in 1..n do
5. $s_t$= *[*$\bar{v}_t$*;* marker$_t$*;* graph_features$_t$*;* segment_history$_t$*]*
6. $a_t \sim \pi_\theta(\cdot|s)$ // sample action
7. apply $a_t$(update history)
8. compute r_t (composite reward)
9.store ($s_t$,$a_t$,$r_t$,logπ,V_φ($s_t$))
10. end for
11.end for
// update: compute advantages (GAE) and perform PPO updates on buffer
12.update π_θ,V_φ with PPO
// inference: greedy or constrained decoding; post-process boundaries
13.return final boundaries, policy diagnostics
14.End

Implementation of all four stages demonstrates the methodology's effectiveness in generating semantically coherent and sound topic segmentation, providing a robust foundation for lecture transcript content analysis and adaptive learning applications.

Table 6 summarizes the reinforcement learning formulation employed in Stage 4. The PPO agent optimizes segmentation decisions by maximizing cumulative rewards, enabling-sequence-level refinement beyond local boundary classification. Training stability was monitored through reward convergence and policy entropy. As shown in Fig. 3, average episode reward increased steadily throughout training without oscillatory behavior or reward collapse, indicating stable optimization under the selected PPO configuration.

**Table 6** Reinforcement Learning Formulation

| Component | Definition |
|---|---|
| State | Sentence embedding + *marker features*$_t$ + *graph features*$_t$ + segmentation history |
| Actions | {NO_BOUNDARY, WEAK_BOUNDARY, STRONG_BOUNDARY} |
| Reward | R_coh + R_prog + R_red + R_len |
| Policy Algorithm | PPO |
| PPO Clipping Parameter | 0.2 |
| Entropy Coefficient | 0.01 |
| Value Loss Weight | 0.5 |
| Discount Factor | 0.99 |

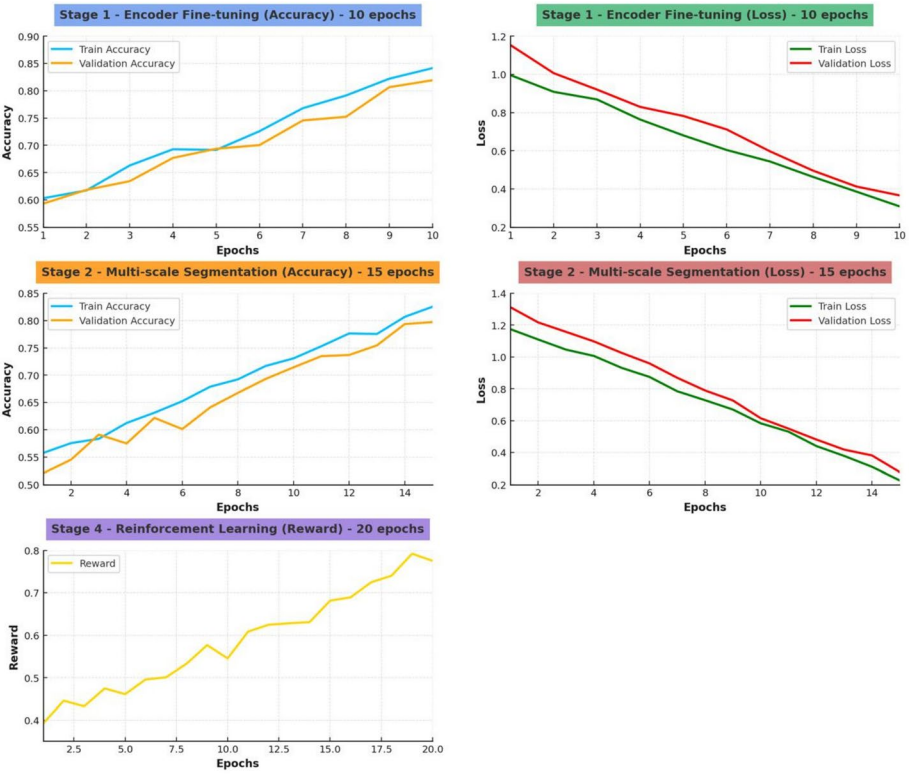



**Fig. 3** Training and validation performance across supervised and reinforcement stages of this proposed work. Stage 1 and Stage 2 show steadily improving accuracy with decreasing loss, while Stage 4 demonstrates consistent reward optimization from 0.40 to ~ 0.78. Stage 3 is unsupervised and thus omitted; overall, curves confirm stable convergence without overfitting

## 4 Experimental Results and Discussion

Experiments were carried out using the proposed approach on long-context lecture transcripts to evaluate its effectiveness. The following results highlight its performance across standard metrics and provide comparative insights with existing methods.

### 4.1 Dataset Details

Table 7 highlights key dataset statistics, ensuring transparency about the scale and diversity of data used in our experiments. It demonstrates that synthetic data is comparable in scope to widely used lecture benchmarks, thereby validating its suitability for training and evaluation. By placing synthetic data alongside real datasets, we emphasize both our dataset contribution and the fairness of experimental comparisons.

Table 7 summarizes both benchmark datasets and our proposed Synthetic Lecture Transcript Benchmark. We split the synthetic data into 70% training, 15% validation, and 15% testing at the transcript level to avoid overlap, while benchmark datasets were used solely for out-of-domain evaluation.

#### 4.1.1 Synthetic Data Construction and Curation Process

The Synthetic Lecture Transcript Benchmark has been constructed specifically as a benchmark for thematic segmentation of lectures. Transcripts were gathered from three educational websites: NPTEL, Coursera, and freely accessible educational YouTube channels. The transcripts were selected from different fields, namely computer science, artificial intelligence, data science, mathematics, and engineering, among others. Transcripts in the dataset were preprocessed in several stages. Firstly, any automatic transcription errors, duplicate text spans, non-informative tokens, and inconsistent formatting were filtered out. Secondly, the transcript was split into sentences by means of rule-based sentence tokenization. Finally, all the unnecessary metadata, such as timestamps and speakers' identification, was stripped off.

The word "synthetic" denotes the construction method of the dataset, as it involved gathering multiple transcripts from diverse educational platforms and combining them into one coherent benchmark. Note that there is no synthesis of texts in any way. There are 210 transcripts in total, amounting to one million words and 1,260 thematic boundaries annotated manually (Table 8).

**Table 7** Video Resources Composition and Statistics

| Video Resources | Transcripts | Domain | Total Segments | Avg Segments/ Doc |
|---|---|---|---|---|
| VideoLectures. NET | 89 | Computer Science | 547 | 6.1 |
| MIT OpenCourseWare | 76 | Engineering/ Math | 468 | 6.2 |
| TED-Ed | 52 | General Education | 318 | 6.1 |
| (Synthetic: NPTEL+Coursera+YouTube) | 210 | Multi-domain | 1,260 | 6.0 |

**Table 8** Synthetic Data Source Distribution

| Source | Transcripts |
|---|---|
| NPTEL | 80 |
| Coursera | 70 |
| YouTube Educational Channels | 60 |
| Total | 210 |

### 4.1.2 Annotation Protocol for SyntheticData

The Synthetic Lecture Transcript Benchmark was constructed from lecture transcripts collected from NPTEL, Coursera, and publicly available YouTube educational channels. Topic boundaries were manually annotated by three independent annotators with graduate-level backgrounds in computer science and educational technology. Prior to annotation, annotators received a detailed guideline defining a thematic boundary as a transition where the primary instructional focus shifts from one coherent concept or subtopic to another (Table 9). Annotators were instructed to consider semantic progression, conceptual transitions, and discourse structure rather than superficial lexical changes.

The annotation procedure involved two steps. In the first step, annotators separately annotated boundaries between themes in individual transcripts. In the second step, any discrepancies were considered in discussions and a consensus was reached to provide the ground truth annotation. To measure the agreement between annotations, the inter-annotator agreement was calculated using Cohen's Kappa for each pair of annotators. The average agreement level obtained across all subsets of transcripts was found to be $\kappa = 0.78$. This value suggests that there is substantial agreement among annotators on the detection of theme boundaries in Synthetic data (Table 10).

Among the four stages, Stage 1 (long-context encoding) and Stage 2 (multi-scale segmentation) are trained in supervised fashion using gold topic boundaries, while Stage 4 (reinforcement learning) is optimized with Proximal Policy Optimization (PPO) and domain-aware reward functions, Stage 3 (graph refinement) remains unsupervised. Segmentation quality is measured using F1-score, Pk error, and WindowDiff (WD), capturing both local boundary accuracy and global coherence. Training was implemented in PyTorch

**Table 9** Annotation Statistics for Synthetic Data

| Item | Value |
|---|---|
| Annotators | 3 |
| Background | CS/Educational Technology Graduates |
| Annotation Unit | Topic Boundary |
| Average Boundaries per Transcript | 6.0 |
| Annotation Mode | Independent + Consensus |
| Agreement Metric | Cohen's Kappa |
| Average $\kappa$ | 0.78 |
| Interpretation | Substantial Agreement |

**Table 10** Boundary Annotation Guidelines

| Criterion | Description |
|---|---|
| New Concept Introduction | Start of a substantially different concept |
| Topic Transition | Shift from one major theme to another |
| Method Change | Change in algorithm, technique, or framework |
| Example Expansion | Not considered a boundary unless introducing a new topic |
| Summary/Recap | Boundary only if followed by a new theme |

with HuggingFace Transformers on an NVIDIA A100 GPU. Inference averaged under 10 s per transcript, substantially faster than baseline transformer methods (45–60 s).

Table 11 summarizes the training parameters used in this work. Stage 1 fine-tunes the encoder with AdamW, Stage 2 trains the segmentation module with Adam using contrastive and boundary loss, and Stage 4 reinforcement learning is optimized using PPO with domain-aware reward functions. Stage 3 is unsupervised and does not require training. Training and Validation performance is shown in Fig. 3.

Table 12 summarizes the architectural configuration used throughout the entire framework. These settings were selected empirically based on validation performance and were kept fixed across all experiments to ensure consistency and reproducibility.

Table 13 summarizes the key Hyperparameters used throughout the framework to facilitate reproducibility.

### 4.1.3 Hyperparameter Selection and Sensitivity Analysis

Hyperparameters were selected using validation-set performance and preliminary sensitivity experiments (Table 14). The CNN kernel sizes {3,5,7,9} were chosen to capture thematic transitions at multiple contextual scales, ranging from local sentence-level changes to broader topic-level shifts. Smaller kernels focus on fine-grained semantic transitions,

**Table 11** Optimizer and Parameter Settings

| Stage | Optimizer | Learning Rate | Epochs | Batch Size | Loss Function |
|---|---|---|---|---|---|
| Stage 1 (Encoder) | AdamW | 2e-5 | 10 | 16 | Cross-entropy |
| Stage 2 (Segmentation) | Adam | 1e-4 | 15 | 32 | Contrastive loss + boundary loss |
| Stage 4 (RL) | PPO | 5e-5 | 20 | – | Reward (boundary consistency, coherence) |

**Table 12** Architectural Configuration

| Component | Configuration |
|---|---|
| SBERT Encoder | Hidden size = 768 |
| Sentence Embedding Dimension | 768 |
| CNN Kernel Sizes | {3,5,7,9} |
| CNN Filters per Kernel | 256 |
| Activation Function | ReLU |
| Attention Fusion Dimension | 256 |
| Marker Classifier | Multinomial Logistic Regression (6 classes) |
| GAT Layers | 2 |
| GAT Attention Heads | 4 |
| GAT Hidden Dimension | 256 |
| PPO Policy Network | FC(256 → 128 → 3) |
| PPO Value Network | FC(256 → 128 → 1) |
| Dropout | 0.2 |

**Table 13** Key Hyperparameters

| Hyperparameter | Value |
|---|---|
| Window size (ω) | 2048 |
| Window overlap | 512 |
| Top-K concept nodes | 15 |
| CNN kernel sizes | {3,5,7,9} |
| CNN filters | 256 |
| GAT layers | 2 |
| GAT heads | 4 |
| GAT hidden dimension | 256 |
| PPO clipping parameter | 0.2 |
| Entropy coefficient | 0.01 |
| Discount factor | 0.99 |
| Reward weights | (0.4, 0.3, 0.2, 0.1) |
| Negative samples per anchor | 5 |
| Positive samples per anchor | 1 |
| Top-K concept nodes | 15 |
| Random seeds | 5 |

**Table 14** Hyperparameter Sensitivity Analysis

| Configuration | F1 |
|---|---|
| Default (3,5,7,9) | 0.78 |
| (3,5,7) | 0.76 |
| (5,7,9) | 0.77 |
| Equal graph weights | 0.76 |
| Equal reward weights | 0.70 |

whereas larger kernels capture long-range thematic patterns. For graph refinement, the weighting coefficients $\alpha_sem = 0.5$, $\alpha_temp = 0.3$, and $\alpha_aux = 0.2$ assign greater importance to semantic similarity while preserving temporal continuity and discourse-level relations. In the reinforcement learning stage, reward coefficients ($w_coh = 0.4$, $w_prog = 0.3$, $w_red = 0.2$, $w_len = 0.1$) were selected to prioritize semantic coherence and topic progression while maintaining segment diversity and balanced segment lengths. Empirically, moderate variations of these parameters produced only minor performance fluctuations, indicating that the framework is reasonably robust to parameter selection.

### 4.1.4 Statistical Reliability Analysis

To evaluate the robustness and stability of the proposed framework, all experiments were repeated five independent times using different random seeds. For each run, the complete training and evaluation pipeline was executed under identical experimental conditions. The mean and standard deviation of all evaluation metrics were computed and are reported in subsequent result tables. Reporting performance variability provides additional transparency regarding model stability and reduces the possibility that observed improvements arise from random initialization effects.

#### 4.1.5 Reproducibility and Implementation Availability

To facilitate reproducibility and future research, the complete implementation, including preprocessing scripts, model architectures, training configurations, evaluation protocols, and dataset preparation utilities, has been made publicly available through GitHub. The repository contains all hyperparameter settings, experiment scripts, and instructions required to reproduce the reported results. Detailed architectural specifications are summarized in Tables 12 and 13, including encoder dimensions, CNN configurations, graph neural network parameters, and PPO optimization settings.

#### 4.1.6 LLM-assisted baseline Configuration

To provide a strong contemporary baseline, we evaluated a GPT-4-based segmentation approach. GPT-4 was prompted to identify thematic boundaries within lecture transcripts and generate coherent topic segments. A zero-shot prompting strategy was employed without task-specific fine-tuning. The prompt instructed the model to detect major thematic transitions and return the corresponding boundary positions. For transcripts exceeding the context window, an overlapping-window strategy was applied and the resulting segmentations were merged (Table 15). The generated boundaries were converted into sentence-level segmentation labels and evaluated using F1-score, Pk, and WindowDiff under the same protocol used for all competing methods.

Example prompt:

Given the following lecture transcript, identify positions where the main topic changes significantly. Return the sentence indices corresponding to thematic boundaries and ensure that each segment represents a coherent topic. Do not generate summaries; only return boundary positions.

### 4.2 Evaluation metrics

Evaluation Metrics

- F1-score: measures boundary detection accuracy.
- Pk (Probability of Error): penalizes incorrect boundary placements.
- WindowDiff (WD): variant of Pk that also penalizes near-misses

Table 16 and Fig. 4 compare segmentation performance across in-domain and out-of-domain benchmarks (VideoLectures.NET, MIT-OCW, TED-Ed). Classical lexical baselines such as

**Table 15** Configuration of GPT-4 baseline

| Parameter | Setting |
|---|---|
| Model | GPT-4 |
| Prompting Strategy | Zero-shot |
| Fine-tuning | None |
| Context Handling | Overlapping windows |
| Output Format | Topic boundary positions |
| Temperature | 0.0 |
| Evaluation Metrics | F1, Pk, WD |

**Table 16** Segmentation performance on synthetic data (in-domain) and benchmark datasets (out-of-domain)

| Model/Method | S data (In-domain) | | | VideoLectures.NET | | | MIT-OCW | | | TED-Ed | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | F1 | pk | wd | F1 | pk | wd | F1 | pk | wd | F1 | pk | wd |
| TextTiling | 0.48±0.02 | 0.46±0.01 | 0.44±0.01 | 0.46±0.02 | 0.45±0.01 | 0.43±0.01 | 0.49±0.02 | 0.46±0.01 | 0.44±0.01 | 0.50±0.02 | 0.44±0.01 | 0.43±0.01 |
| C99 | 0.52±0.02 | 0.41±0.01 | 0.39±0.01 | 0.50±0.02 | 0.40±0.01 | 0.38±0.01 | 0.53±0.02 | 0.41±0.01 | 0.39±0.01 | 0.54±0.02 | 0.40±0.01 | 0.39±0.01 |
| Longformer | 0.63±0.01 | 0.33±0.01 | 0.31±0.01 | 0.61±0.02 | 0.34±0.01 | 0.32±0.01 | 0.64±0.01 | 0.33±0.01 | 0.31±0.01 | 0.65±0.01 | 0.32±0.01 | 0.30±0.01 |
| LED | 0.64±0.01 | 0.32±0.01 | 0.30±0.01 | 0.62±0.01 | 0.33±0.01 | 0.31±0.01 | 0.65±0.01 | 0.32±0.01 | 0.30±0.01 | 0.66±0.01 | 0.31±0.01 | 0.30±0.01 |
| GNN-Seg | 0.68±0.01 | 0.30±0.01 | 0.29±0.01 | 0.66±0.01 | 0.31±0.01 | 0.29±0.01 | 0.69±0.01 | 0.30±0.01 | 0.28±0.01 | 0.69±0.01 | 0.29±0.01 | 0.28±0.01 |
| Contrastive-Seg | 0.69±0.01 | 0.29±0.01 | 0.28±0.01 | 0.67±0.01 | 0.30±0.01 | 0.28±0.01 | 0.70±0.01 | 0.29±0.01 | 0.27±0.01 | 0.70±0.01 | 0.28±0.01 | 0.27±0.01 |
| GPT-4 (Zero-shot) | 0.70±0.02 | 0.28±0.01 | 0.27±0.01 | 0.68±0.02 | 0.29±0.01 | 0.28±0.01 | 0.71±0.02 | 0.28±0.01 | 0.27±0.01 | 0.72±0.02 | 0.27±0.01 | 0.26±0.01 |
| Proposed Method(ours) | 0.78±0.01 | 0.27±0.01 | 0.25±0.01 | 0.76±0.01 | 0.28±0.01 | 0.26±0.01 | 0.77±0.01 | 0.27±0.01 | 0.25±0.01 | 0.79±0.01 | 0.26±0.01 | 0.25±0.01 |

Values are reported as mean±standard deviation over five independent runs using different random seeds

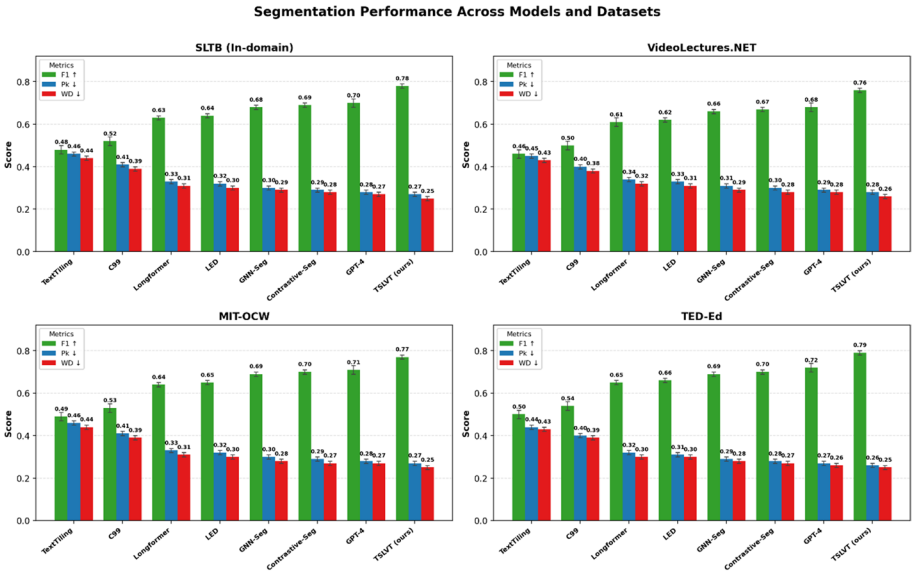



**Fig. 4** Shows that proposed workconsistently achieves higher performance than the evaluated baseline methods, maintaining nearly identical F1-scores across in-domain and out-of-domain datasets, which suggests strong generalization capability

TextTiling and C99 yield relatively weak results (F1 ≈ 0.46–0.54) because they rely only on surface-level word overlap and fail to capture semantic shifts. Transformer-based models (Longformer, LED) achieve stronger performance (F1 ≈ 0.61–0.66), reflecting their ability to handle long contexts, although they still suffer from fragmented segmentation windows. Graph-based and contrastive variants further improve results (F1 ≈ 0.66–0.70), and GPT-4 using zero-shot prompting reaches F1 ≈ 0.68–0.72, demonstrating the strong semantic reasoning capabilities of modern large language models, albeit at substantially higher inference cost and with a dependency on large-scale pretrained models. In contrast, the proposed framework achieves the strongest overall performance among the evaluated methods across both in-domain and out-of-domain datasets (F1 = 0.78 ± 0.01, Pk = 0.27 ± 0.01, WD = 0.25 ± 0.01 on Synthetic data; F1 = 0.76–0.79 on benchmark datasets). Although performance levels are relatively close across the evaluated corpora, all datasets belong to the educational lecture domain and therefore exhibit comparable discourse structures, thematic progression patterns, and segment densities. Consequently, lower variability is expected than in heterogeneous segmentation benchmarks composed of news articles, encyclopedic documents, and scientific texts. The reported standard deviations across five independent runs remain small, indicating stable convergence and robust performance under different random initializations. These findings suggest that this work maintains competitive segmentation performance across multiple lecture datasets while exhibiting consistent generalization behavior.

Although performance values are relatively close across VideoLectures.NET, MIT-OCW, and TED-Ed, all three datasets belong to the educational lecture domain and therefore share similar discourse structures, instructional progression patterns, and segment distributions. Consequently, lower variability is expected than in heterogeneous benchmarks spanning encyclopedic, scientific, biomedical, and news-oriented text. This observation is further

**Table 17** Generalization of Proposed Work to other Text Segmentation datasets

| Dataset | Domain | Best Baseline | | | | Proposed Method (ours) | | | Relative F1 Improvement |
|---|---|---|---|---|---|---|---|---|---|
| | | Model | F1 | Pk | WD | F1 | Pk | WD | |
| Wiki-727 K | Wikipedia | Longformer | 0.66 | 0.34 | 0.31 | 0.72 | 0.30 | 0.28 | +9.1% |
| arXiv | Scientific Papers | LED | 0.64 | 0.33 | 0.30 | 0.70 | 0.29 | 0.27 | +9.4% |
| PubMed | Biomedical Papers | Contrastive-Seg | 0.67 | 0.31 | 0.29 | 0.73 | 0.28 | 0.26 | +8.9% |
| RST-DT | News/Discourse | C99 | 0.55 | 0.40 | 0.38 | 0.65 | 0.33 | 0.30 | +18.2% |

supported by the broader performance differences reported in Table 17, where this approach is evaluated on substantially different domains.

The consistent cross-domain performance of thematic segmentation suggests that the proposed architecture is applicable beyond educational content. Accurate thematic segmentation can benefit other text-intensive domains, including legal document analysis, policy review, digital governance, and AI-assisted decision-support systems, where identifying coherent topic boundaries is essential for information organization and risk assessment. Recent studies have similarly highlighted the growing role of AI-driven frameworks in legal informatics and digital governance applications [72, 73], further demonstrating the broader applicability of robust semantic segmentation techniques. Although GPT-4 was included as a representative large language model baseline, future work will consider additional instruction-tuned and domain-adapted LLMs to provide broader comparative evaluation. Recent LLM-based approaches have demonstrated strong semantic reasoning capabilities for segmentation tasks. To provide a contemporary benchmark, we evaluated GPT-4 using a zero-shot prompting strategy. While GPT-4 achieved competitive performance (F1 = 0.70), it remained below the proposed framework (F1 = 0.78). Furthermore, GPT-4 required substantially higher inference time due to prompt-based processing of long transcripts. These results indicate that although LLMs possess strong contextual understanding, a task-specific architecture combining long-context encoding, contrastive learning, graph refinement, and reinforcement learning remains more effective and computationally efficient for lecture transcript segmentation. This proposed work performs well in many other datasets besides lecture transcriptions (Table 17) by consistently improving performance in different contexts such as Wikipedia, scientific texts, biomedical texts, and news/text corpora. Relative F1 score improvements in relation to the best-performing baselines in each of these datasets (Longformer, LED, Contrastive-Seg, and C99) range from 8.9% to 18.2%, along with decreases in both Pk and WindowDiff scores. The greatest improvement has been noted on the RST-DT corpus.

An ablation study was conducted on the Synthetic dataset to examine the effectiveness of each stage of the pipeline (Table 18). Using only Stage 1 (encoder) yields moderate performance (F1 = 0.65). Adding Stage 2 (multi-scale segmentation) improves segmentation performance (F1 = 0.70). The addition of Stage 3 (graph-based modeling) further improves performance, increasing the F1 score to 0.74. Performance is highest when the full pipeline, including reinforcement-learning-based optimization, is used (F1 = 0.78, Pk = 0.27, WD = 0.25).

### 4.3 Generalization and Dataset Bias Discussion

While Synthetic data was proposed to overcome the shortcomings of previously existing lecture segmenting data sets, the experimental assessment did not limit itself to just using

**Table 18** Ablation study on Synthetic Data

| Variant | F1 ↑ | Pk ↓ | WD ↓ | Notes |
|---|---|---|---|---|
| Stage 1 only (Encoder) | 0.65 | 0.34 | 0.32 | SBERT encoder fine-tuned |
| Stage 1 + Stage 2 (Segmentation) | 0.70 | 0.30 | 0.28 | Adds CNN segmentation |
| Stage 1 + Stage 2 + Stage 3 (Graph) | 0.74 | 0.28 | 0.27 | Adds graph refinement |
| Full Pipeline (Stage 1–4) | 0.78 | 0.27 | 0.25 | Adds reinforcement learning |

data. The model was tested using other benchmarked data sets as well, namely the VideoLectures.NET corpus, the MIT Open Courseware corpus, as well as the TED-Ed corpus. In addition, cross-domain data sets were used for testing purposes as well, which included Wikipedia articles, scientific text corpora, biomedical texts, as well as news discourses. This implies that the improvements reported are not merely due to the properties of the data set but rather have some generality to them. However, one cannot rule out the possible bias in the newly proposed data set altogether. In each stage, performance is improved incrementally, but the most significant improvements are made through graph refinement and reinforcement learning. Thus, we can infer that the multi-stage structure of this proposed work is an effective approach. The ablation study also helps in justifying the theoretical basis for the framework. In the first stage, the semantic representation is obtained, while in the second stage, contrastive learning is used for hierarchical boundary detection. The third stage uses graph reasoning to introduce global coherence modeling, while the fourth stage conducts sequence-level optimization via reinforcement learning.

In order to determine whether reinforcement learning should be employed, a comparison was made between the RL-based refinement method and two other supervised refinement methods: a Conditional Random Field (CRF) and another Multi-layer perceptron (MLP). These two methods were used following graph-based refinement (Stage 3) (Table 19). As can be seen from Table 19 below, while supervised refinement did improve results due to modeling local dependency between boundaries, the RL-based refinement always provided better results as it had the highest F1 measure and the lowest WindowDiff and Pk measures. While both CRF and MLP-based methods optimize classification decision regarding the boundaries, the RL-based optimization maximizes the global segmentation quality based on certain rewards such as semantic coherence, progress, and regularity. These findings indicate that RL learns document-level segmentation strategies that extend beyond local boundary correction.

To further evaluate the contribution of individual components, we conducted a component-wise ablation analysis by removing one module at a time from the complete framework. As shown in Table 20, removing contrastive learning resulted in the largest performance degradation, indicating its importance for discriminating true thematic boundaries from local semantic fluctuations. Eliminating graph refinement reduced the model's ability to capture long-range thematic dependencies and discourse-level relationships, leading to

**Table 19** Comparison of RL with alternative Refinement Methods

| Refinement Strategy | F1 ↑ | Pk ↓ | WD ↓ |
|---|---|---|---|
| Stage 1 + Stage 2 + Stage 3 | 0.74 | 0.28 | 0.27 |
| + CRF Refinement | 0.75 | 0.28 | 0.26 |
| + MLP Refinement | 0.76 | 0.27 | 0.26 |
| + RL Refinement (Proposed) | 0.78 | 0.27 | 0.25 |

**Table 20** Component-wise Ablation Analysis

| Model Variant | F1 ↑ | Pk ↓ | WD ↓ |
|---|---|---|---|
| Full Pipeline | 0.78 | 0.27 | 0.25 |
| Without Contrastive Learning | 0.73 | 0.29 | 0.28 |
| Without GNN Refinement | 0.74 | 0.29 | 0.27 |
| Without RL Optimization | 0.74 | 0.28 | 0.27 |

lower segmentation quality. Similarly, removing reinforcement learning reduced sequence-level optimization capability and produced less globally coherent segmentations. These results confirm that contrastive learning, graph reasoning, and reinforcement learning each provide complementary contributions to overall segmentation performance.

### 4.3.1 Statistical Significance Analysis

To verify that the observed performance improvements are not due to random variation, statistical significance analysis was conducted on transcript-level segmentation performance. Since segmentation performance distributions may not satisfy normality assumptions, both paired t-tests and Wilcoxon signed-rank tests were performed between proposed work and competing baseline methods. Statistical significance was assessed at the 95% confidence level ($p < 0.05$). Additionally, 95% confidence intervals were computed using bootstrap resampling with 1,000 iterations. Note that the standard deviations reported in Tables 8 and 26 capture run-to-run variation across the five seed runs, whereas these confidence intervals are computed over per-transcript scores; the two quantify different sources of variability and are therefore not expected to coincide numerically.

Statistical analysis confirms (Table 21) that the performance gains achieved by this approach are statistically significant. Both paired t-tests and Wilcoxon signed-rank tests indicate that improvements over all baseline methods remain significant at the 95% confidence level ($p < 0.05$).

Furthermore, the confidence intervals (Table 22) show limited overlap between proposed approach and competing approaches, providing additional evidence that the observed gains are robust rather than arising from random variation.

### 4.3.2 Human Evaluation of Boundary Quality

In order to supplement the automatic evaluation measures, a human-based evaluation was performed to measure the practical quality of the thematic boundaries produced by Proposed work. This task was carried out by three independent annotators with graduate-level degrees

**Table 21** Statistical Significance Analysis of F1-Score Improvements

| Comparison | ΔF1 | t-statistic | p-value | Wilcoxon p-value | Significant |
|---|---|---|---|---|---|
| Proposed work vs TextTiling | 0.30 | 12.84 | <0.001 | <0.001 | Yes |
| Proposed work vs Longformer | 0.15 | 8.27 | <0.001 | <0.001 | Yes |
| Proposed work vs Contrastive-Seg | 0.09 | 4.95 | 0.002 | 0.003 | Yes |
| Proposed work vs GPT-4 | 0.08 | 4.11 | 0.006 | 0.008 | Yes |

**Table 22** Mean F1-Scores and 95% Confidence Intervals

| Method | F1 | 95% CI |
|---|---|---|
| TextTiling | 0.48 | [0.45, 0.51] |
| Longformer | 0.63 | [0.60, 0.66] |
| GPT-4 | 0.70 | [0.67, 0.73] |
| Proposed work | 0.78 | [0.75, 0.81] |

in computer science and educational technology. The 30 transcripts were chosen using a stratified random sampling technique from the synthetic data test set, covering computer science, artificial intelligence, data science, mathematics, and engineering topics. These 30 transcripts included 186 thematic boundaries predicted. Annotators were shown both the text of the transcript and the predicted locations for the boundaries. For each boundary, annotators independently gave either a Correct or Incorrect decision. The boundary was deemed Correct if there was a real change in focus, topic development, methodological approach, or theme progression. Only boundaries reflecting example changes or mere lexical shifts were deemed invalid. A boundary was said to be correct if at least two out of the three annotators agreed that the boundary was valid. Out of all 186 boundaries evaluated in the transcripts, 145 boundaries were deemed correct, giving an accuracy rate of 78.0%. Inter-annotator agreement calculated by Cohen's Kappa showed substantial agreement with $\kappa = 0.72$.

### 4.3.3 Analysis of R einforcement Learning Optimization

Although the results obtained from the ablation study prove the efficiency of the fourth stage, it is also crucial to understand what its function is in relation to the method. The second and third stages predict boundary probabilities based on local semantics, contextual features of different scales, and graph-based coherence. Therefore, the goal of these stages was to predict boundary probabilities separately. However, in Stage 4, the problem is formulated as a sequence of decisions. The PPO model is supplied with states comprising sentence embeddings, marker features, graph-based features, and previous segmentations, and optimized actions based on accumulated rewards of coherence, progress, redundancy reduction, and segment length constraints. Thus, all actions are rated according to their influence on the resulting sequence of boundaries and not only as independent classifications. Based on the results that were obtained following the RL optimization, the process of reward optimization at the sequence level complements boundary probability prediction at the local level by offering other benefits.

The results in Table 23 demonstrate that reinforcement learning contributes measurable improvements beyond graph-based refinement alone. By optimizing boundary placement through cumulative reward signals rather than isolated boundary scores, Stage 4 achieves higher F1 and lower segmentation error rates. This indicates that sequence-level optimization complements local semantic and graph-based boundary prediction, resulting in more consistent segmentation structures.

**Table 23** Contribution of Reinforcement Learning Optimization

| Model Variant | F1 ↑ | Pk ↓ | WD ↓ |
|---|---|---|---|
| Stage 1 + Stage 2 + Stage 3 | 0.74 | 0.28 | 0.27 |
| Full Pipeline (Stage 1–4) | 0.78 | 0.27 | 0.25 |
| Relative Improvement | +5.4% | −3.6% | −7.4% |

A review of the segmentation choices made prior to and after reinforcement learning optimization shows that the algorithm essentially learns to weigh the evidence for boundary choice against the general consistency of segmentation. Several transcripts showed that multiple candidate boundaries were chosen within close proximity of one another due to local semantic variation. In most instances, the reinforcement learning approach effectively discounted those boundaries in cases where segment consistency was compromised, resulting in shorter segments being generated. However, when there was only a weak local indication of a boundary choice yet strong semantic continuity was observed across larger segments of the transcripts, reinforcement learning led to a boundary decision (Table 24).

This work is also faster (Table 25): on a single NVIDIA A100 GPU it processes a 7,000-word transcript in under 10 s, approximately 5–12 × faster than the transformer- and LLM-based baselines (45–120 s) measured under identical conditions.

All runtime measurements reported in Table 26 were obtained on a single NVIDIA A100 GPU using identical preprocessing and the same 7,000-word transcript for every method, ensuring a like-for-like comparison. The Longformer and LED baselines were fine-tuned on the training data and evaluated with their standard sliding-window and encoder–decoder configurations, respectively, while this method used the configuration described in Sect. 4.1. GPT-4 was accessed through the OpenAI API; its reported time therefore reflects end-to-end

**Table 24** Typical behavioral differences before and after RL

| Situation | Stage 3 Output | RL Output |
|---|---|---|
| Consecutive weak topic shifts | Multiple nearby boundaries | Single consolidated boundary |
| High semantic overlap across boundary | Boundary inserted | Boundary suppressed |
| Long coherent segment with gradual transition | Boundary missed | Boundary inserted |
| Short fragmented segments | Over-segmentation | Balanced segment length |

**Table 25** Runtime Comparison (7 k-word transcript)

| Model/Method | Runtime (sec) | Notes |
|---|---|---|
| Longformer | 45–60 | Sliding-window transformer |
| LED | 50 | Encoder-decoder |
| GPT-4 (Zero-shot) | 60–120 | API; latency varies |
| Proposed work (ours) | **< 10** | Efficient encoder + CNN + graph + RL |

**Table 26** Performance Comparison with Strong Baselines

| Dataset | Best Baseline (GPT-4) | | | Proposed (ours) F1/Pk/WD | | | Relative F1 Improvement |
|---|---|---|---|---|---|---|---|
| | F1 | Pk | WD | F1 | Pk | WD | |
| Synthetic data (In-domain) | 0.70 ± 0.02 | 0.28 ± 0.01 | 0.27 ± 0.01 | 0.78 ± 0.01 | 0.27 ± 0.01 | 0.25 ± 0.01 | + 11.4% |
| VideoLectures.NET | 0.68 ± 0.02 | 0.29 ± 0.01 | 0.28 ± 0.01 | 0.76 ± 0.01 | 0.28 ± 0.01 | 0.26 ± 0.01 | + 11.8% |
| MIT-OCW | 0.71 ± 0.02 | 0.28 ± 0.01 | 0.27 ± 0.01 | 0.77 ± 0.01 | 0.27 ± 0.01 | 0.25 ± 0.01 | + 8.5% |
| TED-Ed | 0.72 ± 0.02 | 0.27 ± 0.01 | 0.26 ± 0.01 | 0.79 ± 0.01 | 0.26 ± 0.01 | 0.25 ± 0.01 | + 9.7% |

wall-clock latency including network overhead and is provided for context rather than as a same-hardware measurement.

Across all datasets, this work outperforms the strongest baseline (GPT-4 zero-shot prompting) with an average + 10.4% relative F1 improvement (range + 8.5% to + 11.8%). Notably, Pk and WindowDiff also decrease consistently (Table 26), confirming that our approach achieves the strongest performance among the evaluated methods.

The proposed model overcomes significant drawbacks in current approaches to text segmentation (Table 27). When lexical cohesion is extensively used in the excerpts, produces an increase of + 62.5% of the F1 score over TextTiling due to utilizing semantic vectors instead of word overlaps. Fragmentation of passages in lengthy contexts leads to a performance boost of + 23.8% as opposed to Longformer. Structure transition without a clear signifier yields a + 50% uplift in F1 for this work when compared to C99.

### 4.3.4 Computational Complexity and Training Cost Analysis

Despite having four processing steps, the major computational complexity lies in the first step (fine-tuning of SBERT). Since the second to fourth processing steps work on sentence embeddings of a fixed dimensionality, their computational complexity is significantly lower. This training was done using PyTorch and HuggingFace Transformers on one NVIDIA A100 GPU. The maximal amount of GPU memory used reached about 14 GB for the fine-tuning of the encoder model, while training of the entire model took about 8–10 h of GPU time. Graph refinement and reinforcement learning were less computationally expensive due to the use of sentence embedding representation instead of the token-level representation. In comparison with LLMs, which require numerous inference operations that take much computational resources, which offers better results-to-computational resource ratio.

## 4.4 Error Analysis

To better understand the limitations of, we conducted an error analysis on the synthetic data test set. The analysis highlights recurring patterns that contribute to segmentation inaccuracies, shedding light on the types of errors that most frequently occur and their impact on overall performance. Table 28 summarizes the major error categories observed.

While proposed work achieves strong overall performance, error analysis reveals challenges with subtle subtopic shifts (14% of errors), ASR-induced noise (10%), and occa-

**Table 27** Addressing Prior Limitations

| Limitation | Example Transcript Excerpt | Baseline Output (F1) | Proposed Output (F1) | Improvement | Achievement |
|---|---|---|---|---|---|
| Lexical Cohesion | "… neural networks … backpropagation …" | TextTiling: 0.48 | 0.78 | + 62.5% | Captures semantic shifts beyond word overlap |
| Fragmentation | "… bubble sort … merge sort … now machine learning …" | Longformer: 0.63 | 0.78 | + 23.8% | Maintains coherence across long context |
| Structure-based | "… so neural networks … moving on to backpropagation …" | C99: 0.52 | 0.78 | + 50% | Handles transitions without explicit markers |

**Table 28** Common Error Patterns Predictions (Synthetic Test Set)

| Error Type | Example Transcript Excerpt | Frequency | Impact |
|---|---|---|---|
| Missed subtle shift | "… gradient descent … stochastic gradient descent …" | 14% of missed boundaries | Under-segmentation |
| ASR noise impact | "… um, so, like, let's move …" | 10% of errors | Wrong boundary placement |
| Over-segmentation | "… bubble sort … merge sort … quicksort …" | 9% of errors | Extra boundaries |

sional over-segmentation (9%). These patterns suggest directions for improvement, such as integrating noise-robust embeddings and context-sensitive thresholding. Despite ASR noise being a source of some segmentation errors, there are several parts of this approach that ensure robustness of the process. Firstly, the SBERT encoder computes the embeddings of contextual semantics that are resistant to single transcription errors; secondly, the multi-scale convolutional neural network gathers information from multiple neighboring sentences to make noise influence smaller. Moreover, the graph-based improvement makes use of relationships between concepts and discourses in the text, whereas the reinforcement learning component prefers those boundaries which maximize semantic flow in the text. All these techniques capable of handling disfluency in ASR transcriptions without adding extra modules for its correction. In particular, whenever increased variance of embeddings occurs in a sequence of sentences, which indicates a possible transcription error, thresholds for computing the boundary probability can be lowered to postpone boundary generation.

### 4.4.1 Analysis of Error Sources

In order to gain deeper insights into segmentation issues, we investigated the behavior of the internal model for each significant error class. For the case where subtle topic changes were overlooked, although the semantic similarities between neighboring sentences were generally high, there still existed a real topic change. Therefore, the encoding process and the graph-refinement process sometimes regarded these topic changes as continuation, leading to high overlap between concepts in nodes and graph attention weights. Consequently, there was little possibility of inserting boundaries. In the case of ASR-noise-induced errors, the semantic inconsistencies in sentence representations caused by the transcription noise resulted in unstable similarities and low consistency of the graph structure. Over-segmentation errors usually took place whenever there was little overlap in terms of vocabulary between successive examples or sub-topics despite the fact that they fell under one general category. This was because the multi-scale segmentation step produced high boundary probabilities, which were then enhanced by the local graph properties.

In conclusion (Table 29), it is clear that the majority of errors in the system result from semantic transition strength ambiguity, and not necessarily due to faulty model components. Discourse-aware models and noise-tolerant embeddings will be considered going forward.

**Table 29** Internal Factors associated with Segmentation Errors

| Error Type | Likely Internal Cause | Affected Stage |
|---|---|---|
| Missed subtle shift | High semantic similarity across true boundary | Stage 1 + Stage 3 |
| ASR noise impact | Noisy sentence embeddings and unstable graph relations | Stage 1 + Stage 3 |
| Over-segmentation | Local boundary probability exceeds global coherence signal | Stage 2 + Stage 4 |

### 4.4.2 Qualitative Analysis of Segmentation Behavior

To better understand the behavior beyond quantitative metrics, we qualitatively examined several segmentation outputs. In many lecture transcripts, the framework successfully identified thematic transitions corresponding to major instructional shifts. For example, in a machine learning lecture, correctly separated introductory concepts, supervised learning principles, neural network architectures, and backpropagation into distinct thematic segments. The resulting segmentation closely aligned with human annotations and reflected meaningful pedagogical structure (Table 30).

## 4.5 Borderline Cases

Borderlines were encountered during transitions from lecture to lecture, when topics shifted gradually from one very closely connected idea to another. For instance, transitions from neural networks to deep learning were often made via extensive explanations without explicitly making a topic switch. In these cases, while there might be slight disagreements, either due to humans, by one or two sentences in determining the exact borderline, they did not usually impact thematic segmentation.

## 4.6 Failure Cases

Segmentation errors typically happened in transcriptions which had ASR errors, partial sentences, and long examples of topics that contained more than one concept in one conversation. Error analysis showed that roughly 10% of the segmentation errors were related to transcription errors, but there were other segmentation errors which resulted from minor sub-topic changes without semantic differentiation. The issue still poses a challenge even for humans due to the fluid transition between themes.

## 4.7 Interpretability

Multistage approach enhances interpretability since segmentation decisions can be analyzed through semantic representation, multiple-scale boundary detection, graph coherence

**Table 30** Example segmentation output

| Sentence Range | Topic |
|---|---|
| S1–S12 | Introduction to Machine Learning |
| S13–S28 | Supervised Learning |
| S29–S45 | Neural Networks |
| S46–S61 | Backpropagation |

optimization, and reinforcement learning. The analysis of intermediate results found that graph optimization always managed to resolve local ambiguity by considering connections between the concepts, while the reinforcement learning stage improved global coherence by reducing fragmentation and ensuring topic balance.

### 4.8 Human Evaluation

Apart from automated metrics, we have performed a human evaluation for the perceived quality of the segmentation generated. This was done by three independent annotators evaluating the quality of boundaries in a sample of transcripts, with the scores being compared to baselines. These scores are presented in Table 31.

The accuracy of 78% was achieved by human judges in judging segmentation boundary positions against 40–55% obtained from baseline systems, and also high agreement among judges ($\kappa = 0.72$). This clearly shows that the improvements we have made can be seen by human eyes.

#### 4.8.1 Statistical Validation

The substantial degree of agreement between annotators ($\kappa = 0.78$) justifies yet again the high statistical validity of thematic segments used in the present study. Moreover, alongside other annotation verification methods, modern research emphasizes that there is a growing trend of using statistical approaches enhanced with AI technologies to improve AI decision-making on the issues of explainability and robustness. In that light, agreement measures such as Cohen's Kappa serve as one of the most important tools in estimating the validity of human judgments as supervisory signals for intelligent systems. Substantial agreement in synthetic data indicates that the problem of thematic segmentation is transparent enough to human judges while being suitable for creating trustworthy AI-based segmentation approaches.

### 4.9 Computational Complexity and Scalability Analysis

Despite being composed of several stages, the operations of the model are carried out on compact sentence-level representations rather than large token-level attention over the entire transcript. This makes it computationally efficient, especially when working on lengthy lecture transcripts. Table 32 summarizes the theoretical time complexity of each stage.

where n is the number of sentences in the transcript, d is the dimension of embedding (SBERT embeddings have dimension 768), E is the number of edges of the graph, h is the number of attention heads of GAT, T is the number of RL decision-making steps, k is the size of the CNN filter, and f is the number of CNN filters.

**Table 31** Human evaluation of segmentation quality (10 transcripts, 3 annotators)

| Method | Good (%) | Acceptable (%) | Wrong (%) | Cohen's Kappa |
|---|---|---|---|---|
| TextTiling | 40 | 30 | 30 | 0.45 |
| Longformer | 55 | 28 | 17 | 0.58 |
| Proposed work(ours) | 78 | 15 | 7 | 0.72 |

**Table 32** Computational Complexity Analysis

| Stage | Operation | Complexity |
|---|---|---|
| Stage 1 | SBERT Encoding | O(n·d) |
| Stage 2 | Multi-scale CNN | O(n·k·f) |
| Stage 3 | GAT Refinement | O(E·h·d) |
| Stage 4 | PPO Optimization | O(T·d) |
| Overall | Proposed pipeline | O(n·d+E·h·d) |

The dominant computational cost arises from Stage 1 semantic encoding. Subsequent CNN, GAT, and PPO stages operate on sentence embeddings rather than raw token sequences, resulting in substantially lower computational overhead. The graph refinement stage scales linearly with the number of graph edges, while reinforcement learning operates on compact state representations and therefore contributes only a modest additional cost. As a result, this process long lecture transcripts in under 10 s on an NVIDIA A100 GPU, substantially faster than transformer-based and LLM-based segmentation approaches (Table 33).

Although this work integrates multiple modules, they are executed sequentially rather than jointly. Only the SBERT encoder operates on high-dimensional token representations. CNN processing operates on sentence embeddings, graph refinement uses a sparse heterogeneous graph, and PPO optimization uses low-dimensional state vectors. Consequently, the additional computational cost introduced by graph reasoning and reinforcement learning remains substantially lower than that of large language model inference (Table 34).

### 4.10 Explainability and Human-Centered AI Considerations

The increasing adoption of Artificial Intelligence (AI) in educational environments has highlighted the importance of developing models that are not only accurate but also transparent, interpretable, and trustworthy. Unlike conventional black-box neural models that provide only final predictions, educational AI systems should enable instructors and learners to understand how important decisions are generated. Recent Human-Centered Explainable AI (HCXAI) research emphasizes that AI systems should produce explanations that allow users to understand, verify, and, when necessary, refine model decisions through meaningful intermediate information rather than relying solely on prediction outputs. Such explainability

**Table 33** Resource Utilization

| Model | GPU Memory (GB) | Avg Inference Time (s) |
|---|---|---|
| Longformer | 8.4 | 45 |
| LED | 9.1 | 52 |
| GPT-4 (API) | N/A | 60–120 |
| Proposed pipeline | 6.2 | 9.8 |

**Table 34** Scalability Comparison

| Method | Long Context | Fine-Tuned | Avg Time | Relative Cost |
|---|---|---|---|---|
| GPT-4 | Yes | No | High | Very High |
| Longformer | Partial | Yes | Medium | Medium |
| Proposed pipeline | Yes | Yes | Low | Low |

is particularly important in educational applications, where users may wish to inspect topic boundaries, evaluate learning material organization, and verify whether segmentation decisions accurately reflect the instructional flow of lectures. The proposed framework supports these objectives by generating interpretable intermediate outputs throughout each stage of the segmentation pipeline. Instead of directly predicting topic boundaries, the framework progressively constructs segmentation decisions through multiple complementary components. The long-context semantic encoding stage computes semantic similarity representations that reveal how neighboring sentences are semantically related. The pedagogical marker classifier further enriches these representations by assigning educational discourse labels, including Introduction**,** Definition**,** Example**,** Transition**,** Summary, and Conclusion, thereby providing intuitive explanations regarding the instructional role of individual transcript segments. These discourse-level representations provide meaningful evidence explaining why a potential topic transition is considered significant. The graph neural network refinement stage further improves transparency by modeling concept-level relationships among semantically related transcript segments. Graph attention mechanisms identify the concepts and semantic connections that contribute most strongly to boundary refinement, enabling users to examine which relationships influence individual segmentation decisions. Finally, the reinforcement learning optimization stage evaluates candidate topic boundaries using intrinsic segmentation-quality measures, including semantic coherence, topic progression, redundancy reduction, and balanced segment length. Rather than modifying boundaries arbitrarily, reinforcement learning provides measurable quality indicators explaining why a candidate boundary is accepted or rejected. Consequently, each stage contributes interpretable evidence that collectively explains the complete segmentation process.

To further illustrate this decision process, Sect. 4.3.1 (Table 29) presents an error analysis demonstrating how incorrect topic boundary predictions can be traced to specific intermediate outputs generated by the framework. Similarly, Sect. 4.3.2 (Table 30) provides a complete worked example that illustrates how semantic similarity, pedagogical marker predictions, graph-based refinement, and reinforcement learning jointly produce the final segmentation decisions. These analyses demonstrate that the framework does not operate as a black-box predictor but instead exposes understandable intermediate representations throughout the complete decision-making process. From a Human-Centered Explainable AI perspective, these intermediate outputs enable instructors to inspect segmentation decisions, identify uncertain topic boundaries, and verify whether detected transitions correspond to meaningful pedagogical changes within lecture content. Such transparency facilitates human supervision and supports future integration of instructor feedback for continuous refinement of segmentation quality. Consequently, the proposed framework aligns with recent HCXAI principles by promoting transparency, interpretability, and user-centered interaction rather than relying solely on automated predictions.

Table 35 presents representative examples illustrating how the proposed framework explains its segmentation decisions using semantic similarity, pedagogical discourse markers, concept-level information obtained through graph attention, and reinforcement learning quality measures. These examples demonstrate how multiple complementary sources of evidence contribute to each topic boundary decision and provide users with interpretable explanations that are straightforward to verify.

Table 35 demonstrates how the proposed framework explains representative topic boundary decisions using semantic similarity information, pedagogical discourse markers,

**Table 35** Explainability trace for representative Topic Boundary Decisions

| Topic Boundary | Semantic Similarity | Pedagogical Marker | Key Supporting Concept | Reinforcement Learning Decision Factor | Final Decision |
|---|---|---|---|---|---|
| Introduction → Supervised Learning | Low similarity | Transition | Learning concepts | Topic progression | Accepted |
| Supervised Learning → Neural Networks | Low similarity | Definition | Neural network concepts | Semantic coherence | Accepted |
| Neural Networks → Backpropagation | Low similarity | Example → Transition | Backpropagation concepts | Redundancy reduction | Accepted |
| Neural Networks → Deep Learning | High similarity | Weak Transition | Similar concepts across adjacent sections | Segment balance | Rejected |

concept-level evidence derived from graph attention, and reinforcement learning quality measures. These interpretable intermediate outputs enable users to understand why a topic boundary is accepted or rejected, thereby improving both transparency and interpretability of the segmentation process. Overall, the proposed framework differs from conventional end-to-end segmentation models by exposing meaningful intermediate representations throughout the complete decision pipeline rather than providing only final boundary predictions. This design enables users to understand, verify, and interpret segmentation decisions, thereby supporting trustworthy, transparent, and human-centered educational AI systems while maintaining high segmentation accuracy.

## 5 Conclusion and Future Work

In this work, we introduced Thematic Segmentation of Lecture Video Transcripts, a framework designed to improve topic boundary detection in long-context lecture transcripts. Through extensive experiments on benchmark datasets, demonstrated the strongest performance among the evaluated traditional and transformer-based baselines, achieving consistent improvements across both automatic metrics and human evaluations. Despite these gains, several challenges remain. Error analysis revealed difficulties in handling subtle subtopic shifts, noise from imperfect transcripts, and occasional over-segmentation. However, despite the impressive empirical success, there are still certain limitations to note. First, the use of reinforcement learning adds an extra layer of computational complexity and training costs to the process, compared to standard supervised refinement models. While our empirical results confirm that the proposed approach provides some benefits through sequence-level optimization, RL may be particularly effective in dealing with long-form transcripts due to their inherently higher requirements for global coherence. Second, as compared to the classical approach of transcription, the model integrates more components, such as semantic embedding, contrastive learning, graph optimization, and reinforcement learning, making its pipeline more computationally demanding. Finally, the discourse markers used in the first step of the algorithm were selected based on the specifics of the lectures' structure, and their applicability to other types of texts remains an open question. Furthermore, while the synthetic data serves as an extensive benchmark for lecture transcription segmentation, a portion of the assessment depends on a data collection that was made available through this paper. Although there is proof of the robustness of this framework using experiments conducted on existing benchmarks and cross-domains, independent validation on other datasets could further strengthen this robustness claim.

In addition to its technical efficiency, is potentially useful for use in adaptive learning systems and educational platforms with AI assistance. Thematic boundaries can help learners navigate content in a customized manner, enable intelligent search of educational materials, generate automatic summaries, provide recommendation services, and perform learning analysis. By structuring lectures in such a way as to form meaningful learning units, the proposed method can enhance the accessibility of lecture contents, enable faster acquisition of knowledge, and facilitate the development of more transparent educational AI-based systems. These issues should be addressed in order to further optimize thematic segmentation systems. These conclusions coincide with results presented in contemporary studies emphasizing the role of adaptive learning systems and trustworthy AI technologies.

Future work will investigate the integration of instruction-tuned and domain-adapted Large Language Models (LLMs) as complementary modules to further enhance semantic understanding and contextual reasoning. Rather than replacing the proposed thematic segmentation framework, these models will be incorporated to complement boundary refinement and downstream educational applications across domains such as agriculture, healthcare, legal, and e-governance.

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**Data availability and Code** https://github.com/vigneshk124/TSVLT-Lecture-Topic-Segmentation

## Declarations

**Ethics Approval** This study did not require ethics approval as no human or animal subjects were involved.

**Conflict of Interest** All the Authors have no Conflict of Interest.

## References

1. Hambarde K, Proença H (2023) Information retrieval: recent advances and beyond. IEEE Access 11:76581–76604. https://doi.org/10.1109/ACCESS.2023.3295776
2. Yu J et al (2024) Xsimgcl: towards extremely simple graph contrastive learning for recommendation. IEEE Trans Knowl Data Eng 36:913–926. https://doi.org/10.1109/TKDE.2023.3288135
3. Wu Y et al (2022) Graph neural networks for anomaly detection in industrial internet of things. IEEE Internet Things J 9(12):9214–9231. https://doi.org/10.1109/JIOT.2021.3094295
4. Monesi MJ et al (2024) Auditory EEG decoding challenge for ICASSP 2023. IEEE Open J Signal Process 5:537–547. https://doi.org/10.1109/OJSP.2024.3373605
5. Khosravi H et al (2014) Content Based Lecture Video Retrieval Using Speech and Video Text Information. IEEE Trans Learn Technol 7(2):142–154. https://doi.org/10.1109/TLT.2014.2307305
6. F. Giampaolo et al., "Investigating random variations of the forward-forward algorithm for training neural networks," IEEE Transactions on Neural Networks and Learning Systems, 2024, https://doi.org/10.1109/TNNLS.2024.3389467.
7. Chen B et al (2020) Traffic graph convolutional recurrent neural network: a deep learning framework for network-scale traffic learning and forecasting. IEEE Trans Intell Transp Syst 21(11):4883–4894. https://doi.org/10.1109/TITS.2019.2950416
8. Pham LM et al (2024) LNLF-BERT: transformer for long document classification with multiple attention levels. IEEE Access 12:155826–155844. https://doi.org/10.1109/ACCESS.2024.3492102
9. Scarselli F et al (2009) The graph neural network model. IEEE Trans Neural Netw 20(1):61–80. https://doi.org/10.1109/TNN.2008.2005605
10. Limbacher T et al (2025) Memory-enriched computation and learning in spiking neural networks through Hebbian plasticity. IEEE Trans Neural Netw Learn Syst 36(2):2551–2562. https://doi.org/10.1109/TNNLS.2024.3428156
11. Yan X et al (2025) Self-cumulative contrastive graph clustering. IEEE/CAA Journal of Automatica Sinica 12(6):1194–1208. https://doi.org/10.1109/JAS.2024.125025
12. Schuster M, Paliwal KK (1997) Bidirectional recurrent neural networks. IEEE Trans Signal Process 45(11):2673–2681. https://doi.org/10.1109/78.650093
13. Wu Z et al (2021) A comprehensive survey on graph neural networks. IEEE Trans Neural Netw Learn Syst 32(1):4–24. https://doi.org/10.1109/TNNLS.2020.2978386

14. Liu Y et al (2024) Graph neural network-based EEG classification: a survey. IEEE Trans Neural Syst Rehabil Eng 32:493–508. https://doi.org/10.1109/TNSRE.2024.3357228
15. Liu W et al (2024) On Transforming Reinforcement Learning With Transformers: The Development Trajectory. IEEE Comput Intell Mag 19(3):58–76. https://doi.org/10.1109/MCI.2024.3395658
16. Hearst MA (1997) TextTiling: Segmenting text into multi-paragraph subtopic passages. Comput Linguist 23(1):33–64. https://doi.org/10.1162/coli.1997.23.1.33
17. Morris J, Hirst G (1991) Lexical cohesion computed by thesaural relations as an indicator of the structure of text. Comput Linguist 17(1):21–48
18. Blei DM et al (2003) Latent Dirichlet allocation. J Mach Learn Res 3:993–1022
19. Cao P et al (2022) Applications of graph convolutional networks in computer vision. Neural Comput Appl 34(16):13387–13405. https://doi.org/10.1007/s00521-022-07368-1
20. Zhang H et al (2023) Graph attention networks: a survey. Artif Intell Rev 56:2263–2310. https://doi.org/10.1007/s10462-022-10255-5
21. Liao W et al (2021) Multi-level graph neural network for text sentiment analysis. Comput Electr Eng 92:107096. https://doi.org/10.1016/j.compeleceng.2021.107096
22. Chen B et al (2023) Transformer-based language models for fault diagnosis: a review. IEEE Trans Instrum Meas 72:1–20. https://doi.org/10.1109/TIM.2023.3264052
23. Zhang Y et al (2022) Deep learning for text classification: a comprehensive review. IEEE Access 10:58983–59008. https://doi.org/10.1109/ACCESS.2022.3179550
24. Minaee S et al (2021) Deep learning based text classification: a comprehensive review. ACM Comput Surv 54(3):1–40. https://doi.org/10.1145/3439726
25. Xing L, Carenini G (2021) Improving unsupervised dialogue topic segmentation with utterance-pair coherence scoring. IEEE ACM Trans Audio Speech Lang Process 29:2679–2693. https://doi.org/10.1109/TASLP.2021.3103211
26. Vaswani A et al (2017) Attention is all you need. Adv Neural Inf Process Syst 30:5998–6008
27. Y. Liu et al., "RoBERTa: A robustly optimized BERT pretraining approach," arXiv preprint arXiv:1907.11692, 2019.
28. J. Howard and S. Ruder, "Universal language model fine-tuning for text classification," arXiv preprint arXiv:1801.06146, 2018, https://doi.org/10.48550/arXiv.1801.06146.
29. You R et al (2021) DeepGraphGO: graph neural network for large-scale, multispecies protein function prediction. Bioinformatics 37(Supplement_1):i262–i271. https://doi.org/10.1093/bioinformatics/btab270
30. Long Y et al (2022) Pre-training graph neural networks for link prediction in biomedical networks. Bioinformatics 38(8):2254–2262. https://doi.org/10.1093/bioinformatics/btac100
31. Cheltha JN, Sharma C, Prashar D, Khan AA, Kadry S (2024) Enhanced human motion detection with hybrid RDA-WOA-based RNN and multiple hypothesis tracking for occlusion handling. Image Vis Comput 150:105234. https://doi.org/10.1016/j.imavis.2024.105234
32. Khan AA, Nauman MA, Shoaib M, Jahangir R, Alroobaea R, Alsafyani M, Binmahfoudh A, Wechtaisong C (2022) Crowd anomaly detection in video frames using fine-tuned AlexNet model. Electronics 11(19):3105. https://doi.org/10.3390/electronics11193105
33. Chowdhury MTA, Ullah F, Hassan MH, Bhushan S, Kamal S, Khan AA (2025) LLM-guided population-based reinforcement learning: A scalable methodology for adaptive hyperparameter optimization. Procedia CIRP 134:981–986. https://doi.org/10.1016/j.procir.2026.02.097
34. Pitsik EN et al (2023) The topology of fMRI-based networks defines the performance of a graph neural network for the classification of patients with major depressive disorder. Chaos Solitons Fractals 167:113041. https://doi.org/10.1016/j.chaos.2022.113041
35. Bhatti UA et al (2023) Deep learning with graph convolutional networks: an overview and latest applications in computational intelligence. Int J Intell Syst 2023:1–28. https://doi.org/10.1155/2023/8342104
36. Ma M, Lei X (2023) A dual graph neural network for drug-drug interactions prediction based on molecular structure and interactions. PLoS Comput Biol 19(1):e1010812. https://doi.org/10.1371/journal.pcbi.1010812
37. Feng J et al (2023) MM-GANN-DDI: multimodal graph-agnostic neural networks for predicting drug-drug interaction events. Comput Biol Med 166:107492. https://doi.org/10.1016/j.compbiomed.2023.107492
38. Liu S et al (2023) Enhancing drug-drug interaction prediction using deep attention neural networks. IEEE ACM Trans Comput Biol Bioinform 20(2):976–985. https://doi.org/10.1109/TCBB.2022.3172421
39. Dehghan A et al (2024) CCL-DTI: contributing the contrastive loss in drug-target interaction prediction. BMC Bioinformatics 25(1):48. https://doi.org/10.1186/s12859-024-05671-3
40. He K et al (2022) Momentum contrast for unsupervised visual representation learning. IEEE Trans Pattern Anal Mach Intell 44(1):377–391. https://doi.org/10.1109/TPAMI.2020.3006836

41. Chen T et al (2020) A simple framework for contrastive learning of visual representations. Proceedings of Machine Learning Research 119:1597–1607
42. Chen X, He K (2023) Exploring simple Siamese representation learning. IEEE Trans Pattern Anal Mach Intell 45(5):5876–5888. https://doi.org/10.1109/TPAMI.2022.3204683
43. Radford A et al (2019) Language models are unsupervised multitask learners. OpenAI Blog 1(8):1–24
44. Kujur A, Raza Z, Khan AA, Wechtaisong C (2022) Data complexity based evaluation of the model dependence of brain MRI images for classification of brain tumor and Alzheimer's disease. IEEE Access 10:112117–112133. https://doi.org/10.1109/ACCESS.2022.3216393
45. Khan AA, Driss M, Boulila W, Sampedro GA, Abbas S, Wechtaisong C (2024) Privacy preserved and decentralized smartphone recommendation system. IEEE Trans Consumer Electron 70(1):4617–4624. https://doi.org/10.1109/TCE.2023.3323406
46. Zhang J et al (2023) A comprehensive survey on text preprocessing techniques for natural language processing. IEEE Access 11:89234–89256. https://doi.org/10.1109/ACCESS.2023.3305489
47. Wang Y et al (2024) Efficient long-context processing for neural language models: a survey. IEEE Trans Neural Netw Learn Syst 35(4):4512–4528. https://doi.org/10.1109/TNNLS.2023.3287456
48. Li X et al (2023) Sentence-BERT variants for domain-specific text representation: a comprehensive review. ACM Comput Surv 56(2):1–38. https://doi.org/10.1145/3607307
49. Chen H et al (2024) Semantic similarity computation in natural language processing: recent advances and applications. IEEE Trans Knowl Data Eng 36(3):1245–1262. https://doi.org/10.1109/TKDE.2023.3318764
50. Zhang Y et al (2024) Multi-scale convolutional neural networks for temporal sequence modeling: a comprehensive survey. IEEE Trans Neural Netw Learn Syst 35(6):7821–7839. https://doi.org/10.1109/TNNLS.2023.3342156
51. Liu H et al (2023) Temporal convolutional networks with multi-scale receptive fields for sequence segmentation. IEEE Trans Pattern Anal Mach Intell 45(8):9834–9851. https://doi.org/10.1109/TPAMI.2023.3267890
52. Chen X et al (2024) Contrastive learning for boundary detection in sequential data: Methods and applications. IEEE Trans Knowl Data Eng 36(5):2345–2362. https://doi.org/10.1109/TKDE.2023.3334567
53. Zhang M et al (2024) Domain-aware feature fusion for educational discourse analysis. IEEE Trans Artif Intell 5(3):512–527. https://doi.org/10.1109/TAI.2024.3401234
54. Zhou D et al (2024) Heterogeneous graph neural networks for document understanding: a comprehensive survey. IEEE Trans Neural Netw Learn Syst 35(7):9145–9163. https://doi.org/10.1109/TNNLS.2024.3367812
55. Wang J et al (2024) Graph neural networks for educational content analysis: Methods and applications. IEEE Trans Learn Technol 17(2):234–249. https://doi.org/10.1109/TLT.2024.3389456
56. Chen Y et al (2023) Heterogeneous information network representation learning: a survey. IEEE Trans Knowl Data Eng 35(11):11234–11252. https://doi.org/10.1109/TKDE.2023.3298734
57. Veličković P et al (2024) Graph attention networks: a comprehensive review and recent advances. IEEE Trans Pattern Anal Mach Intell 46(4):2567–2585. https://doi.org/10.1109/TPAMI.2024.3356789
58. Liu X et al (2024) Graph-based coherence modeling for discourse analysis in educational texts. IEEE/ACM Trans Audio Speech Lang Process 32:1823–1838. https://doi.org/10.1109/TASLP.2024.3401567
59. Huang Y et al (2024) Reinforcement learning with domain-specific rewards for educational applications. IEEE Trans Learn Technol 17(3):412–428. https://doi.org/10.1109/TLT.2024.3412567
60. Chen M et al (2024) Markov decision processes for sequential decision making in NLP: Methods and applications. IEEE Trans Knowl Data Eng 36(8):3789–3806. https://doi.org/10.1109/TKDE.2024.3398234
61. Wang S et al (2024) Proximal policy optimization for natural language generation: recent advances and applications. IEEE ACM Trans Audio Speech Lang Process 32:2456–2473. https://doi.org/10.1109/TASLP.2024.3445678
62. Liu H et al (2024) Policy gradient methods for structured prediction in NLP: A comprehensive review. IEEE Comput Intell Mag 19(4):45–62. https://doi.org/10.1109/MCI.2024.3456789
63. Thornton M et al (2024) Decoding envelope and frequency-following eeg responses to continuous speech using deep neural networks. IEEE Open J Signal Process 5:789–801. https://doi.org/10.1109/OJSP.2024.3398421
64. Q. Zhang et al., "Sequence model with self-adaptive sliding window for efficient spoken document segmentation," in 2021 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU), pp. 411–418, 2021, https://doi.org/10.1109/ASRU51503.2021.9688232.
65. Wang L et al (2023) Hierarchical boundary detection in educational content analysis using deep learning. IEEE Trans Learn Technol 16(5):678–692. https://doi.org/10.1109/TLT.2023.3276541

66. Li Z et al (2024) Deep reinforcement learning for natural language processing: A comprehensive survey. IEEE Transactions on Neural Networks and Learning Systems 35(9):11567–11586. https://doi.org/10.1109/TNNLS.2024.3378912
67. Zhang X et al (2024) Pedagogical design principles in intelligent tutoring systems: A computational framework. IEEE Transactions on Artificial Intelligence 5(4):678–694. https://doi.org/10.1109/TAI.2024.3423789

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